

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-  
a0b6b5d0c1549870c.jsonl`  
- SHA-256: `779eda65a64f15567a348587b5659e8dee2db86971fa45213e5ab587d01a87dc`  
- Source modified: `2026-06-19T15:49:31+00:00`  
- Imported at: `2026-07-05T16:48:22+00:00`  
- project: `wf_ac0cbb29-6bf`  
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T15:46:34.355Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.

HARD RULES:

- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.

Working dir is the repo root; use Read/Grep/Glob freely.

YOUR SLICE ? status:

Roadmap/status doc staleness contradicted by reality: docs/ROADMAP.md, docs/BACKLOG.md, docs/RALLY_DETECTION_QUALITY_REPORT.md, docs/REAL_FOOTAGE_VALIDATION_REPORT.md, docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md, docs/NEXT_STEPS.md. Look for not-started/todo/planned/pending items that are in fact built and merged (cross-check git history via git log and the code), or completed work still listed as open. Cite the merged PR number or code symbol as evidence. Be conservative: only CLEAR factual status staleness, never roadmap re-prioritisation.

Return EVERY code-contradicting staleness item you can verify in your slice. Exhaustive within the slice; facts only.

2. assistant (2026-06-19T15:46:36.857Z)

I'll audit the roadmap/status docs in my slice for staleness. Let me start by pulling the latest and reading the target docs alongside git history.

3. user (2026-06-19T15:46:38.542Z)

```
ebe2e6d fix(gemini): surface failed chunks loudly (no more silent partial reels)  
a3b4954 feat(runner): --min-rally-duration flag (surface the long-rally reel filter on the CLI)  
(#238)  
a240221 fix(test): ablation litmus uses canonical experiment._is_fusion_schema (unbreaks the  
suite) (#237)  
d749fb9 test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set (#236)  
ea6b2b8 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port  
(#219)  
20f8aa3 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF) (#220)
```

5f555cf refactor(eval): extract one shared golden-feature loader behind the three load_videos (#235)

5409fc4 refactor(api): extract job-worker drain/wake loop from main into backend.api.job_worker (#234)

5c46ebc refactor(eval): split real-fixture subsystem into golden_real_fixtures (pure move) (#233)

97fa187 refactor(db): route bool-returning single-statement writers through _execute_write (#232)

ae10447 refactor(eval): extract _build_preset(args, side) from rally_seg_eval.main() (#231)

1e324ea refactor(motion): extract detection + DB-populate halves from process_video (#230)

53b9cef docs(code-map): point UPLOAD endpoints at backend/api/uploads.py (post #228) (#229)

e67add1 refactor(api): extract chunked-upload island from main into backend.api.uploads router (#228)

8092a39 refactor(segmenter): decompose yolo_hybrid process_video into phase helpers (#227)

f28ce05 refactor(eval): split ablation export_pdf into backend/eval/ablation_pdf (#226)

90dcc57 refactor(gemini): decompose process_video into named private helpers (#225)

18883b5 refactor(stitcher): extract per-segment trim into _trim_one_segment (#224)

af90e32 refactor(db): extract _init_db base-table + per-version migration helpers (#223)

596f7d3 refactor(tracknet): dedup midpoint-split + extract _active_frames helper (#222)

93d7116 fix(config): degrade to defaults + warn when config.json is valid JSON but not an object (#221)

a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)

b460526 docs(recording): assert single-match capture as the #1 rule (now MEASURED) (#218)

ab03043 refactor(eval): single source of truth for the _is_fusion_schema predicate (#215)

ff51b37 chore(eval): re-baseline the nightly after the R1 milestone flip (#204) (#216)

67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)

f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)

db96ddc docs: capture external due-diligence review inputs (research, pick-up-later) (#207)

af5b3e6 docs(rally): tracked doc for closing the detection accuracy gap (golden-driven, per-video) (#211)

55e4c89 docs(archive): LOVO threshold re-sweep at n=13 ? milestone thresholds hold (#214)

bd33085 fix(scripts): unwrap segmenter Result before len() (motion crashes the default eval path) (#213)

c7baf03 fix(db): make v3/v4 migration ALTERS idempotent (interrupted upgrade can brick the DB) (#212)

4bd9d2d fix(eval): ablation litmus tolerates a mixed-schema golden corpus (#209)

be39a70 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17) (#206)

4bbd987 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17)

8736e2d fix(runner): honor stitching watermark/padding + add --detect-from for hi-res reels (#205)

f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)

8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)

2eaccd5 feat(eval): nightly rally-quality regression tripwire (golden corpus, CPU) (#202)

34b9caa feat(eval): R4 ? rally-END inactivity trim (default-OFF) + R3 cap re-scope finding (#200)

4. user (2026-06-19T15:46:38.748Z)

```

1      # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3      This document is the narrative thread: how the indexer evolves from an
4      AI-API-dependent prototype into a self-contained tool that learns rally detection
5      from its own commercially-licensed, annotated video corpus. It explains how we
6      got here, why each step matters, and how it shapes the final tool. Decisions
7      are recorded as ADRs in [DECISIONS.md](archives/decisions/DECISIONS.md); the scoring
mechanics live
8      in [EVALUATION.md](EVALUATION.md).
9
10     > ? Phase 4 update: the authoritative owned-model path is now
11     > [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? a learned
per-frame TAL head
12     > (ActionFormer/ASFormer over WASB trajectory features), not the candidate-window
classifier sketched below.
13     > Audio is parked, not "gated on golden labels": the E1 probe and a follow-up

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foreground-audio test both came back

14 > weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's Gen-1 note).

15

16 ## The thesis

17

18 The indexer detects badminton rallies in long recordings. Its strongest mode

19 (`yolo\_hybrid`) is a two-stage pipeline: cheap **permissive person-detection

20 player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack

21 tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That

22 works, but it costs money, needs the network, and never improves on its own.

23

24 > *Naming note:* the `yolo\_hybrid` registry key is retained for back-compat, but

25 > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +

26 > ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local

27 > player-motion gating stage, not the Ultralytics library.

28

29 A sibling project, **sports-data-collector**, curates sports videos with

30 per-rally `[start, end)` annotations and a commercial-license flag. The thesis of

31 this roadmap: **use that annotated, commercially-usable data to (1) measure the

32 pipeline objectively and (2) train an offline replacement for the paid Gemini

33 stage** ? turning the tool into one that improves as more data is curated, runs

34 air-gapped, and generalizes across sports.

35

36 Why it matters: it removes the per-run API cost and network dependency, gives us

37 a defensible quality metric instead of eyeballing, and creates a flywheel ?

38 more curated matches ? a better offline segmenter ? better highlights.

39

40 ## The four phases

41

42 ### Phase 1 ? Bridge the repos (DONE)

43 A sport-agnostic `curate export` in the collector emits the indexer's `start,end`

44 ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on

45 the `is\_commercial` flag. ? *Why:* one clean producer/consumer contract that works

46 for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.

47 See **ADR-002**.

48

49 ### Phase 2 ? Acquire full-resolution video (DONE)

50 The corpus was 256x144 (the original crawl used `worstvideo`). The `fetch` command

51 re-acquires every video at **best available resolution (1080p here)**, **in place**

52 (preserving annotations), via **authenticated Firefox Premium cookies** and yt-dlp's

53 `default` client. ? *Why:* YOLO/shuttle detection needs real pixels; and the

54 journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate

55 limiting) now encoded in the tooling. **22/22 fetchable matches are now 1080p.**

56 See **ADR-003**.

57

58 ### Phase 3 ? Evaluate + tune (IN PROGRESS)

59 `scripts/run\_phase3\_eval.py` walks the manifest, processes each video, and scores

detected

60 rallies against the ground truth ? per video and corpus-aggregate (precision/recall/

61 F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final`

62 (refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error

63 lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation**

64 problem (candidates catch them, refinement is loose). That diagnosis decides where

65 effort goes, and establishes the baseline any trained model must beat.

66

67 ### Phase 4 ? Learn an offline, API-free segmenter (NEXT)

68 Replace the paid Gemini refinement with a **lightweight classifier trained on the

69 rally annotations**, consuming features the pipeline already computes and stores

70 (`candidate\_segments.stats`). No external API, no new labeling. ? *Why:* this is the

71 payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from

72 data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate

73 stage at **recall ~0.05 @ IoU 0.5** on `shuttleset\_8` (median best coverage of a

74 rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's

75 substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

```

76
77 ## Complementary local cues (beyond the segmenter swap)
78
79 Once the substrate is WASB shuttle-motion, the next lever is fusing additional
80 fully-local cues to raise rally recall/precision without reintroducing a paid
81 API. The first one explored is audio "thwack" hit-detection (ADR-007,
82 accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in
83 `backend/pipeline/audio/`. The E1 GO/NO-GO probe has run on real GoPro audio ?
84 verdict AMBER / leaning-negative: recall signal is real (hit-clusters recover
85 WASB-missed rallies) but the classical-onset precision ceiling (~2.2x) is too weak
86 to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a
87 learned timbre classifier ? but per the owned-model plan audio is now parked (the
88 foreground-audio idea also failed), not revived now that labels exist.
89 See `archives/research/AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is the more
credible #2 cue.
90
91 The recurring blocker across Phases 3?4 and the audio work is the same: more
92 golden labels is the single highest-leverage move (see the cross-cut note below).
93
94 ## How this shapes the final tool
95
96 - Offline & free by default. The production rally mode becomes a local model;
97 Gemini is kept only as an upper-bound reference, not a dependency.
98 - Self-improving. Every newly curated, annotated match is more training signal;
99 re-training is a local, deterministic step.
100 - Measurable. Every change is scored against the golden set, so "better" is a
101 number, not a vibe.
102 - Sport-generic. The label ? feature ? classifier machinery and the export
103 bridge already span sports; adding a sport is data, not new architecture.
104 - Commercially clean. Only `is_commercial` data flows through the bridge, so the
105 trained model and any derived product rest on a licensable foundation.
106
107 ## Status at a glance
108
109 | Phase | What | Status |
110 |-----|-----|-----|
111 | 1 | Export bridge (`curate export` ? manifest + CSVs) | ? Done |
112 | 2 | 1080p acquisition (`fetch`, in-place, cookies) | ? Done (22/22) |
113 | 3 | Eval + tuning loop (`scripts/run_phase3_eval.py`) | ? In progress |
114 | 4 | Offline learned segmenter (WASB shuttle-motion substrate) | ? Next (substrate
decided: WASB, ADR-004) |
115 | + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1 probe run ?
AMBER (~0.6 AUC), parked / de-prioritized |
116
117 > Cross-cutting: golden-set ground truth (which powers Phases 3?4 and on-demand
regression testing)
118 > is produced via a vendor process + pluggable annotation provider ? see
119 > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) and ADR-006. As of 2026-06-02
the annotation
120 > guideline is authored (collector `docs/annotation/`) and Roora (Bengaluru) is
selected for an
121 > initial 3-sport pilot (badminton, pickleball, table tennis). Input-capture quality
requirements
122 > that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
123 >
124 > Observability: every processed video now emits an always-on per-video diagnostics
125 > report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
126 > pipeline regressions and footguns surface without re-running eval ? see
127 > [CODE_MAP.md](CODE_MAP.md) §2 (reporting/observability).
128

```

5. user (2026-06-19T15:46:39.258Z)

```

1 # Next steps (live) ? owned-model + multisport

```

```

2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is
CPU-only: `torch-cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30     ## Where we are (one paragraph)
31     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
32     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
33     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor

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34     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
35     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
36     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
37     battery is done** (see table) ? the moat is quantified.
38
39     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
40     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
41     |---|---|---|
42     | Heuristic (velocity windowing) | 0.657 | 0.157 |
43     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
44     | Two-stage WASB+Gemini refine | 0.83 | ? |
45     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
46
47     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
48     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
49     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
50     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
51     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
52     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
53
54     ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
55     **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
56
57     **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records).** See archived
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
58
59     - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
layer) .
60     `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
61     `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md`
(2-layer mental model).
62     - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
implements the net-crossing **"Gate A" (`apply_gate(..., min_crossings=2)` that kills
service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
`min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter
specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists
`net_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)
63     - **Current status (2026-06-14):** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
64     - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO):**

```

65 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)

66 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).

67 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

68 - ****Discipline (from EXPERIMENT_HARNESS.md):**** every new cue ****flag-gated, default
OFF****; promote to default ****only on measured LOVO lift**** through `run\_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

69 - ? Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion
paths are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL
machine).

70 - ****External due-diligence review inputs (2026-06-17) ? PICK UP AFTER** current ideas are
exhausted; full details + citations in `RALLY_QUALITY_RESEARCH.md`
§5.6.****** An owner-commissioned 31-reference review of the quality report ****validated**** the
methodology and surfaced these ****future**** levers (research inputs, not active work, not new
claims; verify citations before any submission):

71 1. ****"First-mile" paper framing (highest-value):**** stroke-classifiers (BST/DSTA) + the
big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all ****assume a pre-trimmed
rally clip already exists**** ? we solve the ignored prerequisite. Pitch the paper as ****the
missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage.****

72 2. ****Ground R2 in Action-Spotting / Precise-Event-Spotting****
(SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.

73 3. ****Add domain baselines for any submission:**** MonoTrack + BST on ShuttleSet / Fine-
Badminton (we have only heuristic + Gemini today).

74 4. ****Prefer TemporalMaxer over ActionFormer**** for the M0.4 scorer swap (parameter-free
max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

75 5. ****Court polygons ? pose ? the rally-END lever:**** masking on-court players can
resolve ****occluded end-of-rally states**** (the #1 remaining gap vs Gemini), not just spectator
noise.

76 6. ****Name the eval?serve lesson "pipeline distribution skew"**** ? a citable paper
"lesson."

77

78 ****Discipline reminder:**** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, ****then**** advance.

79

80 **## Prioritized next steps**

81 1. ****[owner, in progress] GROW THE GOLDEN CORPUS**** ? the single highest-leverage move;
every lever below feeds on it.

82 Priority order for new videos: ****a) multi-court badminton**** (the target distribution
AND where the ship target

83 lives), ****b) ?3 matches per new sport ? table tennis first**** (WASB transfers at F1
0.909; pickleball labels are

84 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per

85 ``VIDEO_RECORDING_GUIDELINES.md``; ****adult sessions only for the training pool****
(consent guardrail). Per video:

86 label ? ``curate import-local --normalize --license Proprietary --fps <true fps>`` ?
re-run the Gen-0 harness.

87 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.

88 2. ****[owner decision] Set the ship-gate target**** ? now informed: floor = heuristic 0.480
(necessary), ****multi-court**

89 reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93

90 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@IoU0.5 ? 0.70
on multi-court folds with

91 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

92 3. ****M0.4 next increment (\$0/local):**** swap the Gen-0 LogReg for a lightweight temporal
head (MIT, minutes on the

93 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the

94 LOVO-vs-oracle gap is visible per fold in the harness output). ****Prefer
TemporalMaxer**** (parameter-free

```

95         max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review
96         (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
97     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
98         badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
99         by C3 (provenance multiplies per sport).
100    5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
101        "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
102        (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
103        `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
104        physical-visit approach; reframed product-feedback-first per owner spec):**
105        `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
106        `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
107        rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108        capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
109        decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
110    6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
111        deletion in AI Studio); **rename repo off "badminton"** (owner action,
112        in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114    ## Pointers / artifacts
115    - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
116    Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
117    `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118    - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119    manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120    - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
121    (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
122    labels: `...\Collect\Golden Labelled\`.
123    - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124    (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125    with a MISLEADING "credits depleted" 429; ramp gradually.
126    - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127    PLATFORM slice = async job model (single-tenant).
128

```

6. assistant (2026-06-19T15:46:42.766Z)

This is a CPU-only dev container; I'll work read-only and not attempt to run the GPU pipeline. Let me read the remaining target docs and check git history for relevant items.

7. user (2026-06-19T15:46:43.714Z)

```

1    # Backlog ? deferred items to revisit
2
3    Single tracker for work we consciously deferred, so nothing gets lost between sessions.

```

```

Each item records what, why it's deferred, and the trigger to revisit. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5     > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to GitHub issues instead (or in addition).
6     > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8     _Last updated: 2026-06-10._
9
10    ---
11
12    ## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
`NEXT_STEPS.md`)
13    - ? Rename the repo off "badminton" ? owner-emphatic "ASAP" action; the project is
multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. Revisit:
owner action, before broad external sharing.
14    - ? Quantify cost-to-Gen-2 (the largest open risk) ? benchmark real VRAM/token/$ for
the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 678 GPUs)
before standing up any owned serving stack. Revisit: before Phase-2 commitment.
15    - ? Define the ship-gate F1@tIoU target ? "very high P/R" must be operationalized
(floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). Revisit: at M0.4
harness.
16
17    ## Engineering hardening deferred from the 2026-06-10 audit sweep
18    - ? App-factory / lifespan + lazy torch-cv2 imports ? `uvicorn backend.main:app`
leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
(#16). Revisit: when the async slice starts.
19    - ? WSL temp/frame/cache + Gemini remote-file GC ? staged videos, PNG frame dumps,
`*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
Revisit: before any volume/batch processing.
20    - ? shlex-quote WSL config values ? DONE 2026-06-11 (#97):
`detectors/base.py::wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
tilde expansion coexist; applied to every `~`-capable arg in both runners;
`tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
(repo_dir/conda_env) if config ever becomes per-tenant.
21
22    ---
23
24    ## Monetization / licensing
25    *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
26
27    - ? Ultralytics Enterprise pricing ? get a quote only if the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently not
needed (swap done). Quote-only pricing. Revisit: if detector quality forces reconsidering
YOLOv8.
28    - ? Codec patent thresholds (AVC/HEVC/AAC) ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds with legal counsel before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. Revisit: before commercial launch.
29    - ? Per-checkpoint weight/dataset licenses ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the code license ? the distributed weights/dataset terms.
Either train our own or confirm each checkpoint. Revisit: when adopting any borrowed
weights.
30    - ? Real GPU $/video & latency ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. Revisit: when choosing the production LLM step.
31
32    ## Detector swap follow-ups
33    *(source: PR #8, ADR-005)*
34
35    - ? `supervision` ByteTrack deprecation ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and removed in 0.30.0; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTracker` from the separate `trackers` package before lifting the

```

cap. **Revisit:** before bumping supervision past 0.30.

36 - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but **not yet validated on actual footage** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.

37 - ? **yolo_hybrid** naming ? the registry key, `yolo_*` config keys, and `yolo-hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a clean break is worth the migration.

38

39 **Golden-set subproject**

40 `*(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*`

41

42 - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief + illustrated guide in the collector repo `docs/annotation/`` ? rally CSV schema; ignore dead time / include every rally (incl. faults & lets) / shot = one contact; `ending_reason`` (forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for badminton & pickleball.

43 - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the vendor answer key. **Unblocked.** **Revisit:** anytime.

44 - ? **GS-3** ? `HumanVendorProvider`` ? concrete adapter for the chosen annotation vendor (submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora** (Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the Roora pilot confirms quality, then build the adapter.

45 - ? **GS-4 oracle model** ? `AutomatedProvider`` interface is stubbed (this work) with a deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it **MUST** be a different model lineage than production, or the regression test is blind to shared failures. **Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).

46 - ? **GS-5** ? regression triggers ? wire `regress()` into CI (pre-merge gate), on-demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path. **Revisit:** after GS-4.

47 - ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the \$4 bar; escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as calibration once the gold key exists.

48 - ? **Phase-1 video collection** ? gather the round-1 golden corpus per `docs/GOLDEN_SET_PHASE1_VIDEOS.md``: **static self-recorded** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with **? held out by match**. **Unblocked.**

49 - ? **Train/eval separation guardrails** (was `DEFERRED_HARDENING_PLAN`` #13; design preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs** today, kept apart by convention only; nothing *enforces* that a video used to train the learned segmenter (`backend/eval/training.py::build_training_set`` ? `distill_local``) is absent from the eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design** (from the archived DEFERRED plan §2): split unit = **match**, never clip (hold ~? of each sport as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md``); add a `split: "train"|"eval"`` field to the collector's `curate export`` manifest (cross-repo contract, ADR-002); new `backend/eval/splits.py`` with `load_split(manifest) -> (train_ids, eval_ids)`` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in `compute_video_id``) + `assert_disjoint(train_ids, eval_ids)`` that raises on overlap; wire the guard into the training entrypoint (`distill_local`/`training``) and into `run_regression.py`` (baseline computed strictly on the eval split). **Verification:** unit test (overlapping ids ? `assert_disjoint`` raises) + integration (a deliberately-leaked manifest fails the training entrypoint). Full rationale/risks: [`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md``](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2. **Revisit:** dedicated design session with the user (needs a collector-side manifest change).

50

51 **Recording best practices**

52 `*(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*`

53

54 - ? **Validate recording hypotheses** ? empirically test which capture conditions (resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote the winners into user-facing "how to record" guidance. Only the static-camera rule is currently evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure per-condition deltas.

55

```

56     ## Code & test organization
57     *(source: owner review comments on PR #35, 2026-06-07)*
58
59     - ? tests/` reorganization to reduce fragmentation** ? owner wants to isolate test
components on a grouping
60     metric (by subsystem/component) so tests/` is less fragmented, mirroring the new
backend/pipeline/audio/` +
61     backend/eval/audio/` layout (e.g. audio tests grouped under tests/audio/`, eval
tests under tests/eval/`).
62     The exact grouping metric is the owner's design choice** (consolidate small files
vs. add per-area dirs ? the
63     "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
**dedicated refactor PR**,
64     not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The
backend/eval/audio/` half of
65     the request ? comment 1 ? was done in PR #35.)
66
67     ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)
68
69     *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
second-in-command)*
70
71     - ? Docs-only checkpoint PR** ? This update + the refreshed
docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (since archived)
formally record the approved path before any code changes. The initial review docs (PR #37)
introduced the findings; this entry locks the execution decisions.
72     - ? Approved directory structure** ? Adopt a layered approach under backend/` for
professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-
file dirs):
73     - backend/config/` (package ? start with this)
74     - backend/api/` (FastAPI app, routes, static UI)
75     - backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
76     - backend/infrastructure/` (database + external adapters)
77     - Keep backend/eval/` prominent as a first-class concern.
78     - Logical grouping for providers, sports, annotations, utils, and a small core/` only
when needed.
79     - ? Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1
(Config) **first**, introducing the backend/config/` package immediately. Perform the full
directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes
conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
80     - ? Deliverable 1 (Config)** ? backend/config/` package (models.py` + loader.py`)
instead of a single config.py`. Typed AppConfig`, central loader with built-in defaults + file
override, extra="allow" for migration. Update setup.py` + main.py`. Add tests. Land this
before the layout move.
81     - ? Subsequent items** ? See the refreshed docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging,
packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its
own high-pri step right after Config.
82
83     **Status legend reminder:** ? = approved / in progress tracking, ? = do when triggered.
84
85     ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
86     *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
#11/#12; shared lib obsreport`)*
87
88     - ? Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are
DONE.** Remaining = items 3?9 in docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived).
89     - ? Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a
CUSTOMER-friendly summary), via the **soft-dependency** lib obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes <video>.report.json /
.report.md / .summary.md` next to the video.
90     - ? Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage
+ footgun flags + verdict already surface quality; a clean customer summary already exists).
**Do NOT build a parallel mechanism** ? instead EXTEND the report: add an audio-readiness

```

```

stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is
picked up.
91   - ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider
timing.** The report records AI timing as `not_captured`; once `wasb_hybrid`/`tracknet_hybrid`
are unified there is ONE place to thread a recorder callback through `provider.analyze_video` ?
the report. **Revisit:** with item 7 (Claude to add the timing hook there).
92   - ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report
INFERS failures (e.g. `silent_provider_noop` = 0 segments + no API key). A structured
Result/error from providers/segmenters would let the report read the real reason as a first-
class flag. **Revisit:** with item 3.
93   - ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via
editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos
once SSH/CI access to the private lib repo exists.
94   - ? **`run_all_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL,
pytest exit code) before sending PRs.
95   - ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the
two typed-config bugs from
96       #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2)
**`output_dir` is a real field on
97       **`OutputConfig`** (default "output"); the CLI overrides it on the typed model in
`main()`, retiring the
98       write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5 derivation
into
99       `backend/utils/hashing.py::compute_video_id` and migrated all four `run_*.py` scripts
off raw `json.load` onto
100      `backend.config.load_config`. See `CODE_MAP.md` §2.0/§6 (footguns retired).
101

```

8. user (2026-06-19T15:46:44.427Z)

```

1      # Local, Cheap, AI Rally Detection for Racket Sports ? Quality Report (Checkpoint)
2
3      **Internal · honest measured results only · checkpoint 2026-06-17 · supersedes the
2026-06-14 draft**
4      **Project: badminton-highlight-indexer · toward a paper**
5
6      > *An honest empirical study of a shuttle-trajectory rally segmenter, its over-
segmentation fix, a
7      > task-faithful evaluation metric, a disciplined small-N promotion gate, and a first
real-footage
8      > validation against human ground truth and a cloud VLM.*
9
10     ---
11
12     ## Abstract
13
14     We present a fully local, low-cost rally-detection system for racket sports (badminton
first) that
15     segments a match video into rally `[start, end]` intervals. The system is built on a
**frozen
16     shuttle-trajectory substrate** (a WASB CNN that emits per-frame shuttle position), from
which a
17     velocity signal seeds recall-oriented candidate windows that a tiny pure-numpy logistic-
regression
18     scorer grades. Every additional cue enters as a confidence-weighted **feature column**
the scorer
19     learns to weight ? never a hand-coded branch ? and ships **default-OFF**, promoted only
on a measured
20     held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches**
(grown from
21     n=6 this cycle, including **5 real owner-hand-labelled ground-truth clips**), treating
the labelling
22     vendor as an interchangeable oracle: the contract is with the golden labels, not their
provenance.
23

```

24 Since the last checkpoint (2026-06-14) the headline result moved from "one promoted-then-reverted cue"

25 to a **statistically significant over-segmentation fix**. Bounded windowing plus a rally-END

26 inactivity trim (the "cap=6 + R4" milestone) lifts our **task-faithful tolerance-match** F1 from 0.091

27 to 0.323** on n=11 (? +0.232, 95% CI [+0.153, +0.304], paired sign-test **10/0**, p=0.002**), drives the

28 **merge-rate** from 0.694 to 0.150** (rallies-swallowed-by-a-merge, down on **all 11** videos), and

29 tightens the mean rally-END error from **52.1 s** to **3.8 s**. A first ground-truth comparison against a

30 cloud VLM (Gemini) shows local reaching **~86% of Gemini on rally detection** (F1@0.25) with the

31 rally-**START** already at parity** ? the entire residual gap is the rally-END. Negative and null

32 results are reported in full (person and audio fusion cues remain non-significant at n=6; a

33 net-crossing gate that wins offline does **not** transfer to the served operating point). We also

34 introduce a deliberately honest **task-faithful evaluation metric** (asymmetric tolerance-match with

35 strict 1:1 assignment) after finding that plain temporal-IoU is the wrong ruler for this task in both

36 directions. The contribution is less any single number than a **method for being honest** at small

37 scale** ? and the path to a publishable result is now a short one: more labelled matches.

38

39 ---

40

41 ## Headline results at a glance

42

43 | | |

44 |---|---|

45 | **Over-segmentation milestone** (n=11, R2 tolerance-F1) | **0.091 ? 0.323** . ? +0.232 . CI [+0.153, +0.304] . sign-test **10/0**, p=0.002** |

46 | **Merge-rate** (golden rallies swallowed by a merge) | **0.694 ? 0.150** (down on all 11 videos; p=0.001) |

47 | **Mean rally-END error** | **52.1 s ? 3.8 s** |

48 | **Real-footage ground truth** (n=2)** | local **~86 % of Gemini** on detection (F1@0.25); rally-**START** **at Gemini parity** |

49 | **Gemini cloud reference** | clean **0.93** / multi-court **0.66** (the owned-model ship target is the 0.66) |

50 | **Golden corpus** | **n=11 matches** (grown from 6), incl. 5 real ground-truth clips; minors excluded; no biometric labelling |

51 | **Promotion status** | all cues/tiers **default-OFF**; the milestone config flip is owner-gated |

52

53 ---

54

55 ## 1. Introduction & Problem

56

57 A rally is the unit of play in racket sports: the span from serve contact to the point ending.

58 Indexing a long match video into its rallies is the substrate for everything a coach or player wants ?

59 highlight reels, "show me the longest rally," per-rally statistics. Cloud vision-language models do

60 this well on clean footage, but they are expensive per video, non-deterministic, network-bound, and ?

61 critically ? **cannot be a training-data source for an owned model under their terms**. Our constraint

62 is the opposite end of the design space: a system that runs **locally, cheaply, reproducibly** on a

63 single consumer GPU, with paid LLMs allowed only as an optional inference fallback,
never in the
64 training loop.
65
66 Two failure modes dominate and motivate the method:
67
68 1. **Held-shuttle false positives** ? a player holding or flicking the shuttle between
points produces
69 shuttle/player motion the old velocity heuristic counted as a rally (the owner-
observed precision bug).
70 2. **Over-segmentation / over-merge** ? the segmenter treats "shuttle is moving" as "a
rally is in
71 progress," so dead time and serves **bridge several true rallies into one mega-**
window (the
72 Tier-0 baseline merged **52 % of golden rallies**; `kushagra_singles` was **2 windows**
for 32 rallies*).
73
74 Crucially, the standard temporal-IoU metric is **nearly blind to over-merge** (an
overlap-based match
75 absorbs it), so a method can look fine on IoU while destroying rally structure. Both
failure modes are
76 about the **boundary between play and non-play** ? exactly what this report's windowing
tiers and
77 fusion gates target, and exactly why we had to build an over-segmentation-aware metric
before we could
78 measure the failure we set out to fix.
79
80 ---
81
82 **## 2. System & Method**
83
84 **### 2.1 A two-layer mental model**
85
86 **Layer 1 ? Detection** ("where is the shuttle this frame?"). A frozen WASB CNN emits a
per-frame
87 `(x, y, visible)` shuttle track. The shuttle is the camera-invariant primary signal
because "shuttle in
88 flight" is a near-synonym for "a rally is happening," unlike player motion (which fooled
the old
89 heuristic on broadcast-pan cameras).
90
91 **Layer 2 ? Segmentation** ("when is a rally?"). From the per-frame track we compute a
velocity signal
92 that seeds recall-oriented candidate windows; a learned scorer grades each window into
final
93 `[start, end]` rallies. The chain is: `frames ? WASB frozen trajectory ? velocity ?`
shuttle-seeded
94 `candidate windows ? windowing tiers ? learned scorer over feature columns ? graded`
rallies`.
95
96 A key honesty property follows from the data: golden labels are rally **times** (when),
not `(x, y)`
97 coordinates (where). They therefore **cannot retrain the frozen WASB detector** ? they
can only grade
98 and tune the segmentation layer. The discipline is "calibrate on some rallies, measure
on held-out ones."
99
100 **### 2.2 Signals as feature columns, not branches**
101
102 Every producer (detector / analyzer) emits a `(value, confidence)` signal; the scorer
learns to weight
103 those columns and there are no hand-coded `if/else` branches in the decision path. The
scorer is
104 deliberately a tiny pure-numpy **L2 logistic regression** (ADR-004), not a neural net,
so it survives

105 small-N. A cue that does not earn its keep simply stays at weight ? 0 ? it costs nothing because it is

106 an unused column, not a live branch. Fusion has two realizations: a **decision-level gate** (label-free,

107 ships first) and a **feature-level path** (the learned route), where cues emit small,

108 scale/translation-invariant per-window features (counts, ratios, zone-membership, two-sidedness ? never

109 raw coordinates, which would memorize this court and camera).

110

111 **### 2.3 The windowing tiers (this cycle's quality levers)**

112

113 All are **default-OFF**, config-gated, and only ever **cut** windows (recall cannot drop):

114

115 - **W1 ? bounded windowing.** Any window longer than a cap is recursively split at its largest internal

116 inactivity gap (the See-et-al. sustained-inactivity boundary). Fixes the over-merge.

117 - **R3 ? cap re-scope (12 s ? 6 s).** The rally upper bound; a finer cap measured strictly better.

118 - **R4 ? rally-END inactivity trim.** After a rally the shuttle keeps drifting slowly (picked up /

119 walked), over-extending the window END; R4 trims the post-rally tail back to the last sustained-active

120 frame. START is left untouched (it is already accurate).

121 - **W1.5 / R5 ? hard-cap & blob-fallback.** Last-resort midpoint chops for ***continuously-active*** clips

122 with no inactivity gap to split on (a degenerate ~174 s blob). Surgical, do-no-harm, default-OFF.

123 - **W2 ? net-crossing rally-state gate.** Prunes windows whose shuttle never crossed the net (the

124 held-shuttle / service-feed discriminator).

125

126 ---

127

128 **## 3. Evaluation Methodology**

129

130 This section is the heart of the work: the system's credibility rests on **how** it is measured.

131

132 **### 3.1 The owned golden corpus and the oracle contract**

133

134 Evaluation uses an owned golden corpus, **grown from n=6 to n=11 this cycle** by adding 5 real

135 owner-hand-labelled ground-truth clips (``mahadevpura_1/_2``, ``GX010128``, ``GX030094``, ``Badminton_BXH_2``)

136 to the original six (``adarsh_avi_singles``, ``kushagra_singles``, ``largetest_doubles``, ``gbaaddy``, ``testlarge_short``,

137 ``mahadevpura_singles``; 262 rallies). The corpus is classified Proprietary, **minors excluded entirely**,

138 no biometric/identity labelling. The labelling vendor (Roora) is treated as an **interchangeable oracle**:

139 the contract is with the golden labels ? trustable ground truth validated by automated checks plus a

140 ~20 % human spot-check at ± 0.3 s boundary tolerance ? not with their provenance. Evaluation rigor is

141 therefore decoupled from the vendor, and the corpus scales by adding or swapping oracles. **This is the**

142 project's moat: a consented dataset plus an honest eval harness, not the weights.

143

144 **### 3.2 Leave-one-video-out, with honest statistics**

145

146 Learned variants are scored **leave-one-video-out (LOVO)**, grouped by match: train on the other videos'

147 candidates, predict the held-out video, score against its labels ? never a held-out ***window*** from the

148 same video (windows within a video are correlated and flatter the score). Because a bare LOVO mean at

149 small N is not trustworthy, every promotion attaches: a **bootstrap 95 % CI**, a **paired ?F1 CI**, and

150 an **exact paired sign test**. The promotion rule is encoded directly: **a lift is real only when the**

151 **?F1 CI clears 0 and the sign test agrees.**

152

153 **### 3.3 A task-faithful metric (R2) ? tIoU was the wrong ruler**

154

155 We found plain temporal-IoU-F1@0.5 is the **wrong ruler in both directions**: for a ~6 s rally it

156 tolerates only ~±1.9 s, and the intrinsically-fuzzy ~1.5 s service window nearly exhausts that

157 (over-penalizing the START); and its overlap-based matching is murky about **over-production** (W1

158 over-segments ~1.5× yet tIoU hides it via overlap). We therefore introduced **R2**, a task-faithful

159 metric: an **asymmetric tolerance-match** (``|?start| ? ?_start = 2.0 s` AND `|?end| ? ?_end = 1.5 s``)

160 under **strict 1:1 Hungarian assignment** (never greedy overlap), reporting ``P@? / R@? / F1@?``, a

161 **start/end boundary-error split**, an **over-segmentation guard** on normalized merge/split rates, and

162 a **?-sweep** {1.5, 2.0, 2.5, 3.0}. **tIoU/mIoU** are demoted to secondary diagnostics. **R2** is honest by

163 construction: it **lowers** our headline number (tIoU-F1 0.440 ? tolerance-F1 0.323 on the milestone) by

164 charging for the over-production and loose ends tIoU absorbed.

165

166 **### 3.4 Over-segmentation-aware harness + dual-eval-set discipline**

167

168 ``segmentation_metrics.merge_split_report`` breaks predictions into matched / **merge** (1 pred ? 2

169 rallies) / split / missed / spurious, surfacing ``merge_rate` / `split_rate`` ? the failure tIoU hid.

170 A **nightly regression tripwire** (CPU-only; WASB trajectories are cached, only the windowing/scorer

171 re-runs) re-scores the golden corpus under two configs ? **DEFAULT** (served production windowing, all

172 milestone knobs OFF) and **MILESTONE** (cap=6 + R4) ? and trips on any gated regression. The owner-adopted

173 **dual-eval-set discipline** maintains a growing GOLDEN-finalized eval set + a RAW eval set so that a

174 late win cannot silently undo an earlier one.

175

176 **### 3.5 Persist once, re-score forever**

177

178 Expensive, non-deterministic per-frame detection (WASB trajectory, person tracks, audio hits) runs once

179 per video on the GPU and is persisted as a small replayable feature substrate. The cheap, deterministic

180 decision (windowing, gates, scorer) then re-scores any number of **Variants** offline, with no GPU and

181 zero production risk ? the reproducibility thesis of the whole harness.

182

183 ---

184

185 **## 4. Results**

186

187 **### 4.1 Reference baselines that drive strategy**

188

189 Two reference footages anchor strategy: **Mahadevpura** (clean-ish, single court) and **Kushagra**

190 (multi-court, the hard case). The reading is blunt: clean footage is **commoditized** (a

Gemini serve

191 reaches 0.93), but multi-court drops the cloud wrapper to **0.66 ? and that 0.66 is the
owned model's

192 ship target**, not the clean ceiling.

193

194 **Table 1 ? Rally-detection F1 at IoU 0.5 vs human golden labels (per-venue
references).**

195

196	Method Mahadevpura (clean) Kushagra (multi-court)
197	--- --- ---
198	Heuristic (velocity windowing) 0.657 0.157
199	Gen-0 owned learned head (LOVO, n=6) 0.744 0.561
200	Two-stage WASB + Gemini refine 0.83 ? (not measured)
201	Gemini wrapper (`gemini-flash-latest`) **0.93** **0.66**

202

203 The owned Gen-0 head already reaches **0.561** multi-court at n=6 ? a corpus-growth-
sized gap to the

204 0.66 target, not an architecture-sized one. (The aggregate **heuristic LOVO floor**
across n=6 is

205 **0.480**; the learned head's aggregate LOVO is 0.626, sign-test 5/6, **p=0.109 ? not
yet significant,

206 ship=False**.)

207

208 ### 4.2 The over-segmentation milestone (this cycle's headline)

209

210 Bounded windowing (W1), a finer cap (R3, 6 s), and the rally-END inactivity trim (R4)
compose into the

211 **"cap=6 + R4" milestone**, measured on the full **n=11** corpus under the task-faithful
R2 metric.

212

213 **Table 2 ? Windowing ladder (n=11, R2 tolerance-F1, ?_start=2.0 s / ?_end=1.5 s).**

214

215	Config tolerance-F1 (mean [95 % CI]) tIoU-F1 (secondary) mean \ ?end\
216	--- --- --- ---
217	baseline (all-OFF) 0.091 [0.026, 0.179] 0.195 52.1 s
218	+ W1 cap=12 0.168 [0.100, 0.241] 0.357 6.5 s
219	+ cap=6 (R3) 0.276 [0.174, 0.380] 0.407 5.7 s
220	**+ R4 (milestone)** **0.323 [0.210, 0.433]** **0.440** **3.8 s**

221

222 **Table 3 ? Promotion evidence (paired sign-tests).**

223

224	Comparison ?tolF1 95 % CI sign-test significant?
225	--- --- --- ---
226	baseline ? milestone **+0.232** [+0.153, +0.304] **10/0/1, p=0.002** ?
227	cap=12 ? cap=6 (R3) +0.108 [+0.070, +0.154] 10/0/1, p=0.002 ?
228	cap=6 ? +R4 +0.047 [+0.022, +0.072] 8/0/3, p=0.008 ?
229	W1 alone (n=6, tIoU-F1, served) +0.139 clears 0 6/0, p=0.031 ?
230	W2 net-crossing gate (n=6, tIoU-F1, served) +0.019 [+0.011, +0.027] 6/0, p=0.031 ? (golden) / not GT-backed

231

232 The milestone is the **first statistically significant rally-quality win of the
project** and ? unlike

233 the earlier Gate-A episode (\$4.5) ? it is **served-measured** (the milestone *is* a
served windowing

234 config), so there is no eval?serve transfer risk.

235

236 ### 4.3 The over-merge is fixed (the failure tIoU hid)

237

238 The over-segmentation-aware metric shows the pathology directly and its repair:

239

240 - **merge-rate 0.694 ? 0.150** on n=11 ? down on **all 11** videos (? ?0.545, sign-test
0/11, p=0.001).

241 (On the original n=6 served view, W1 alone took merge-rate **0.523 ? 0.038**.)

242 - **mean rally-END error 52.1 s ? 3.8 s** ? the mega-window collapse is gone.

243 - The **net over-segmentation guard** (a weighted merge/split-rate cost, merges weighted

higher because

244 a merge hides a rally = recall loss) **passes** (net cost 1.388 ? 0.423), confirming the structural

245 change reduces real over-segmentation rather than trading it for over-splitting.

246

247 **### 4.4 First real-footage validation against ground truth**

248

249 The capstone of the cycle: the windowing work measured on **6** real Kushagra/Yash clips** (fully-local,

250 no-AI WASB, with Gemini as a strong reference) and ? for the first time ? against **owner-hand-labelled**

251 ground truth** on two of them. The over-merge fix is **real at scale** (aggregate over 6 real clips:

252 windows 48 mega-windows ? 250; mean window length 62.9 s ? 10.8 s; merge-groups 27 ? 10; mean

253 matched-IoU 0.23 ? 0.39). The signature pathology ? `GX010129`: *2 windows / 267 s / 27 rallies

254 trapped* ? is gone. W1 then mildly over-corrects into over-segmentation (~1.5x), and we verified this is

255 **clip-content-driven, not venue-driven** (the "GoPro-vs-court" story is unsupported).

256

257 **Table 4 ? First ground-truth at-par (vs human-labelled rallies).**

258

Clip (golden rallies)	metric	Gemini (cloud)	best LOCAL (no-AI)
---	---	---	---
mahadevpura-2 (40)	F1@0.5 / F1@0.25	0.557 / 0.835	0.296 / 0.722
GX010128 (52)	F1@0.5 / F1@0.25	0.813 / 0.857	0.457 / 0.756

263

264 Local is **~86%88 %** of Gemini on rally detection (F1@0.25)** but only **~53%56 %** on tight boundaries

265 (F1@0.5)**. On GX010128 local even **out-recalls Gemini** ? it overlaps all 52 golden rallies; Gemini

266 matched only 39/52 (missed 13).

267

268 **Table 5 ? The gap is the rally-END; the START is already at Gemini parity (matched-pair median, s).**

269

Clip	local \?start\	Gemini \?start\	local \?end\	Gemini \?end\
---	---	---	---	---
mahadevpura-2	1.74	1.46	3.12	1.19
GX010128	0.78	0.98	2.12	0.76

274

275 Local **nails the START** (? or better than Gemini); the **entire residual gap is the rally-END**

276 over-extending. This crisply localizes the next lever: a stronger sustained-inactivity rally-END rule,

277 **not** a generic boundary head and **not** a serve/START classifier (START is solved). **Honest caveat:**

278 mahadevpura-2 hit 1.03x of Gemini's rally **count** yet F1@0.5 only 0.150 with W1+W2 ? right count, wrong

279 boundaries. **Count parity is a deceptive proxy; ground truth is the only honest judge.**

280

281 **### 4.5 The Gate-A eval?serve finding (reported in full)**

282

283 A net-crossing gate (`net\_crossings ? 2`) was promoted offline (?F1 **+0.074**, 6/6, p=0.031, CI

284 [+0.042, +0.104]) and briefly flipped to the served default ? then **reverted**. On the served **coarse**

285 windowing the gate is **neutral-to-negative** (mean served ?F1 ?0.008, ~12 real rallies dropped)**. The

286 root cause is canonical: the offline harness used a recall-oriented **proposal**

287 windowing that

287 over-segments (so the gate had abundant junk to cut), whereas the served windowing already merges

288 aggressively (so the gate trims ****real**** rallies instead). The honest, served replacement is ****W2****

289 (§4.2): on the now-bounded W1 windowing the same net-crossing signal is a small but robust ****+0.019**

290 (6/0)****** win ? a satisfying reversal that confirms ***the cue's value is windowing-dependent***. A standing

291 litmus (``served_gate_a.py``) pins this behaviour and trips if a future change silently revives or breaks

292 it. ****We do not count the reverted +0.074 as a live win.****

293

294 **### 4.6 Fusion ablation: person and audio (null / uncertain at n=6)**

295

296 GPU person-feature extraction finished all 6 original golden videos; the LOVO drivers measured the lift

297 with honest stats. Neither cue clears the bar (Table 6) ? both stay ****default-OFF****.

298

299 ****Table 6 ? Fusion ablation (n=6, honest LOVO, IoU 0.5).****

300

Variant	F1	?F1 vs prior	95 % CI	sign-test	clears 0?
shuttle-only	0.307	? ? ? ?			
+ person (P3)	0.286	?0.021	[?0.165, +0.162]	2W/4L, p=0.688	No
+ person + audio (P4)	0.366	+0.079	[?0.054, +0.203]	4W/2L, p=0.688	No

306

307 Person adds noise (without court polygons, the in-court count includes spectators). Audio is the more

308 promising lead (wins 4/6; +27.7 % point estimate) but n=6 cannot confirm it. ****We report these**

309 **deliberately**** ? negative and uncertain results are the honest ablation backbone.

310

311 **### 4.7 The learned-variant operating-point fix**

312

313 An early window-level logistic variant lost to control and zeroed a video ? a fixed-0.5-cutoff artifact,

314 not a real defect. Adding an additive, default-OFF ****threshold-tuning step** (operating point chosen on

315 the train fold)****** plus probability calibration (Platt / isotonic) lifted mean F1 ****0.307 ? 0.423****

316 (+0.116), recovered ``testlarge`` from 0.000 to 0.300, and the learned variant now beats CONTROL (0.388).

317 The lesson ? ***calibrate and tune the operating point inside the LOVO loop*** ? is now baked into the harness.

318

319 ---

320

321 **## 5. Detailed comparison with Gemini**

322

323 Gemini (``gemini-flash-latest``) is our ****reference and optional inference fallback ? never a training**

324 source, never the owned target on clean footage******. The comparison is multi-dimensional, and "who wins"

325 depends entirely on the axis.

326

327 ****Table 7 ? Owned local system vs Gemini cloud wrapper.****

328

Dimension	Gemini wrapper (cloud)	Owned local system
Accuracy, clean (Mahadevpura)	**0.93** F1 (P 1.00 / R 0.87 / mIoU 0.80 / boundary 0.7 s)	Gen-0 0.744; windowing milestone closes most of the over-merge
Accuracy, multi-court (Kushagra)	**0.66** F1 (P 0.69 / R 0.62 / mIoU 0.73 / boundary 1.2 s)	Gen-0 **0.561** ? 0.10 behind, corpus-sized
Rally **detection** (F1@0.25, ground truth)	reference	**~86?88 % of Gemini** ; out-recalls it on GX010128 (52/52 vs 39/52)
Rally **START** boundary	parity	**at parity** (<code>`\ ?start\ `</code> 0.78?1.74 s vs 0.98?1.46 s)

335 | Rally ****END**** boundary | tight (\\|?end\\| ~0.8?1.2 s) | the ****gap**** (\\|?end\\| 2.1?3.1 s) ? the next lever |

336 | Cost | reference | ****< \$1** for both reference videos****** at flash pricing (serial, ~14 chunk requests); ****\$0**** for the local path |

337 | Determinism | ****non-deterministic**** (VLM; outputs vary run-to-run) | ****fully deterministic**** ? re-scoring yields byte-identical intervals |

338 | Reliability | ****503** capacity waves throughout****** (one chunk needed ~5 min backoff); a cold/prepaid project burst-throttles with a misleading "credits depleted" 429 | local; no external dependency. ****Sole-provider** flakiness is why the provider-registry fallback is load-bearing****** |

339 | Privacy / offline | requires upload + server-side processing + live network | ****100 % local****; CPU-only for the decision layer after one GPU pass |

340 | Training-data licensing | ****barred**** ? paid-LLM outputs can never train the owned model (terms/copyright) | owned corpus + vendored labels only; ****clean MIT/Apache/BSD**** weights path |

341

342 ****The strategic reading.**** Clean footage is ****commoditized**** ? for a few cents a Gemini call returns

343 0.93, so we **serve** clean footage with Gemini and reserve owned weights for where they matter. The

344 multi-court ****0.66**** is the line the owned model must reach to be economically justified; the Gen-0 head

345 is already at 0.561 with only n=6. Two-stage WASB+Gemini refine (0.83 clean) is the cost-efficient

346 serving path for long tapes (it uploads only candidate clips). The honest bottom line: ****Gemini wins on**

347 absolute boundary accuracy today; the owned system wins on cost, determinism, privacy, offline operation,

348 and the ability to be trained and owned ? and it has already reached START parity and ~86 % detection

349 parity on real ground truth.******

350

351 ---

352

353 **## 6. Limitations**

354

355 - ****Sample size.**** n=11 (5 with real ground truth) ? enough for direction-checks, LOVO sign-tests, and

356 now significant windowing results, but not enough to claim a general model. Corpus growth is the

357 highest-ROI unblock.

358 - ****Person & audio CIs span 0**** at n=6; neither fusion cue is promotable yet.

359 - ****Court polygons absent**** ? the mechanical reason +person is a wash (counts spectators).

360 - ****Served path partially live-verified.**** The over-merge windowing is served-measured; the full fusion

361 segmenter on live video is verified offline on cached trajectories, and the milestone config flip stays

362 owner-gated.

363 - ****Detector domain gap unmeasured.**** WASB/TrackNet and the person detector were trained on other rigs;

364 their accuracy on our amateur/static-tripod footage is not yet quantified, and every downstream cue

365 inherits that gap.

366 - ****C3 ? WASB checkpoint provenance**** is a commercial-ship blocker a clean head does not launder; an

367 owned or cleanly-licensed detector is required before commercial release.

368

369 ---

370

371 **## 7. Planned work ? next steps & expected wins**

372

373 1. ****Grow the corpus via the oracle (the #1 lever).**** Priority: multi-court badminton (the target

374 distribution), then ?3 matches per new sport (table tennis first ? WASB transfers at

F1 0.909 in the literature). **Expected win:** at n=10 with 9/10 wins the exact sign-test reaches **p ? 0.011** ? and we are already at n=11 with 10/0 on the milestone (p=0.002), so the cue-level results become publishable as the corpus and ablations widen.

2. **Promote the milestone (cap=6 + R4) to the served default** ? a 2-line config flip, owner-gated; the significant, served-measured, no-regression win of this cycle.

3. **Close the rally-END gap** ? the ground-truth-localized lever (|?end| 3.8 s ? toward Gemini's ~1 s): strengthen the sustained-inactivity END rule; R4 already bought ?1.9 s (+0.047, 8/0, p=0.008).

4. **Activate court polygons.** The intake manifest already collects `court\_corners / net\_line / singles\_or\_doubles`; masking off-court persons is the direct fix for the person-cue spectator noise.

5. **Re-measure person/audio on the bounded windowing** (the GPU golden-feature LOVO), and measure audio on shuttle-only **directly** (without the person handicap; the E1 probe found ~2.2× discrimination).

6. **ActionFormer/ASFormer (MIT) swap for the Gen-0 logistic head** ? \$0, minutes on a 2070S. *Caveat from the literature:* parameter-free pooling beats learned 1-D conv at small N (TemporalMaxer), which is exactly why the current scorer is a logistic regression ? so this is gated on corpus growth.

7. **Owned VLM (Qwen3-VL, Apache-2.0; InternVL3 MIT fallback)** ? TAL-head-first (boundary accuracy is the product; fine-tuned VLMs miss strict boundaries), commercial-clean across code, data, and weights.

8. Areas yet to be explored

- **Detector domain gap** ? quantify WASB/TrackNet and the person detector on our own footage (false negatives, drift); every cue inherits this unmeasured gap.
- **Served-path live end-to-end** ? run the fusion segmenter on a real static-tripod clip and confirm the held-shuttle/warm-up/spectator false-positive drop directly (not only on cached trajectories).
- **Two-stage refine on multi-court** ? measured at 0.83 clean, **unmeasured** on the multi-court target distribution where the moat lives.
- **Continuously-active footage** ? R5 rescues the degenerate blob (0.000 ? 0.238) but blindly; a rally-state-aware split (net-crossings / two-sided motion) is the principled fix.
- **Pose / stance (parked)** ? Apache-2.0 options exist but domain shift on court angles is unproven; box-first remains the baseline.
- **Formal active learning** ? *which* video to label next (max local-vs-Gemini disagreement + scorer uncertainty + venue/fps diversity) is designed but not yet a method.
- **Multisport clean detectors** ? table-tennis WASB transfer is unverified on our footage; pickleball needs a clean MIT/Apache ball detector.

9. Related work & path to submission

Related work (cite, don't claim). The architecture's skeleton is standard and is cited, not claimed.

416 Candidate-windows-then-learned-scorer is **temporal action localization** (proposal-then-score). The
 417 over-segmentation fixes are **temporal action segmentation** refinement / post-processing. "A learned
 418 arbiter over cue columns" is **late / score-level fusion** realized as stacked generalization (? 20 years
 419 old). In-domain prior art (**See et al. 2024/25** rally start/end via sustained inactivity ? which W1/R4
 420 directly port; **Hsu et al. 2024** trajectory+swing hit detection; MonoTrack; ShuttleSet) hard-codes the
 421 same cues as ``if/else`` verdicts. **(Citations are reused from the project's own adversarially-verified
 422 literature review; none are introduced unverified here.)**
 423
 424 **What is genuinely novel ? and narrow.** (1) The **task-faithful asymmetric tolerance-match metric**
 425 with strict 1:1 assignment and an over-segmentation guard, motivated by tIoU's measured blindness for
 426 this task; (2) the **report-first reconciler-as-linter** + the **disciplined small-N promotion gate**
 427 (default-OFF cues, bootstrap CI + paired sign test, promote-only-when-CI-clears-0, rollback = config
 428 flip). Everything else (the fusion mechanism, the persist-once substrate, the no-regression harness) is
 429 textbook, and we claim only the specific discipline.
 430
 431 **Honest publishability verdict.** This is an **applied / systems** contribution, not an algorithmic one
 432 ? there is no new loss, architecture, or dataset family. At the previous checkpoint (n=6, one
 433 promoted-then-reverted cue, two null cues) it was **not** conference-ready. This cycle changes that
 434 materially: a **significant** over-segmentation result at **n=11** ($p=0.002$), a **task-faithful metric**
 435 contribution, and a **ground-truth-backed START-solved / END-gap diagnosis**. The minimal additions for a
 436 credible submission are concrete and within reach:
 437
 438 - **Corpus ? n ? 16**, with multi-court isolated as its own held-out slice (don't conflate clean and
 439 multi-court baselines).
 440 - **Lead with the tolerance-match metric + over-segmentation metrics** (merge-rate, F1@k), not bare tIoU.
 441 - **Per-cue ablation with CIs** on the grown corpus (the ``ablation.py`` harness already exists).
 442
 443 **Target venues** (applications/systems-oriented): **CVIU**, **ICIP**, **IEEE Trans. Multimedia**, or a **CVPR/**
 444 **ICCV workshop** (CV-for-sports / fine-grained video). Not the CVPR/ICCV main track (no algorithmic
 445 novelty). The cheapest path to a publishable result remains **more labelled matches**, not more
 446 engineering.
 447
 448 ---
 449
 450 **## 10. Engineering & reproducibility**
 451
 452 - **Experiment leaderboard** ? every ``compare()`` appends one reproducible row to
 453 ``eval_baselines/experiment_leaderboard.json``, pinned to ``(spec_hash, label_set_hash, IoU)``, so any
 454 regression traces to a recipe change, not drift.
 455 - **Nightly regression tripwire** ? CPU-only (cached trajectories), guards **DEFAULT** (served) and **MILESTONE**
 456 configs; exit 1 = regression, 4 = stale baseline.

```

457     - Standing litmus ? `served_gate_a.py` pins the honest eval?serve facts.
458     - CI gates ? Ruff, lenient-but-green mypy with a ratchet, syntax+import smoke (no
GPU stack), and the
459         full offline mocked suite with a coverage floor. Combined with default-OFF cues + a
pinned CONTROL
460         variant, rollback is a config flip, never a `git revert`.
461     - Privacy-safe fixtures ? golden-regression fixtures + de-attribution tooling allow
video-free
462         shipping of the eval sets.
463
464     ---
465
466     ## 11. Conclusion
467
468     The contribution of this work is less any single F1 number and more a method for being
honest at small
469     scale ? and this cycle shows that discipline paying off. On an owned golden corpus
grown to n=11
470     (now with real ground truth), bounding the windowing and trimming the rally-END produced
the project's
471     first statistically significant rally-quality win (tolerance-F1 0.091 ? 0.323,
p=0.002), drove the
472     merge-rate down on every video (0.694 ? 0.150), and ? measured against human ground
truth for the
473     first time ? put the rally-START at parity with a cloud VLM, localizing the entire
remaining gap to
474     the rally-END. We reported the nulls in full (person, audio), banked the eval?serve
lesson (the
475     reverted Gate-A ? the honest W2), and introduced the task-faithful metric that made the
over-production
476     tIoU hid finally visible. The architecture stays deliberately standard where the
literature is mature; we
477     claim novelty only on the task-faithful metric and the small-N promotion discipline. The
path to a
478     publishable result is now short and concrete: more labelled matches, multi-court
isolated, ablations on
479     the grown corpus ? the moat was always the consented data plus the honest harness, not
the weights.
480

```

9. assistant (2026-06-19T15:46:47.398Z)

Now let me read the remaining two docs and check the current config defaults, since several claims hinge on whether the milestone (cap=6 + R4) is still "owner-gated/default-OFF" or has been flipped ON.

10. user (2026-06-19T15:46:48.888Z)

```

1     # Local (No-AI) Rally Detector ? Real-Footage Validation & Improvement Roadmap
2
3     Date: 2026-06-16 . Audience: owner/cofounder . Status: validation complete,
roadmap proposed (all flags default-OFF)
4     Footage: 6 Kushagra/Yash June-12 clips, 1080p proxies, fully-local no-AI WASB
windowing vs Gemini (strong reference, not ground truth except mahadevpura-2, owner hand-
labeled).
5     Tracked doc to fold this into: `docs/GOLDEN_REGRESSION_FIXTURES.md` . root-cause
backbone: `docs/RALLY_QUALITY_RESEARCH.md` §2.
6
7     ---
8
9     ## 1. Headline verdict
10
11     The W1 `max_window_duration` cap fixed the over-merge catastrophe ? aggregate count
went from 48 (29% of Gemini's 165) to 250 (152%), and mIoU rose 0.23 ? 0.39 ? but it overshot
into over-segmentation and never landed near 1.0, so we traded a merge problem for a split

```

problem rather than solving localization. On the one clip with owner ground truth (mahadevpura-2), the best local config is ****~86% of Gemini at loose tolerance (F1@0.25 0.722 vs 0.835) but only ~53% at tight boundaries (F1@0.5 0.296 vs 0.557)**** ? and the entire residual gap is ****rally-END over-extension**** (|?end| 3.12s vs Gemini 1.19s), while rally-START is already near Gemini parity (|?start| 1.74s vs 1.46s).

12

13 ---

14

15 ## 2. What worked ? the over-merge fix at scale

16

17 The baseline collapsed entire games into mega-windows. The cap demolished that pathology:

18

19 | Metric (aggregate, n=6) | Baseline | W1 (cap=12) |

20 |---|---|---|

21 | Rally count | 48 (0.29× Gemini) | 250 (1.52×) |

22 | Mean window length | 62.9s | 10.8s |

23 | Merge-groups | 27 | 10 |

24 | mIoU | 0.23 | 0.39 |

25

26 - The signature pathology ? ****GX010129: 2 windows, mean 267s, 27 Gemini rallies trapped in one merge, mIoU 0.0**** ? is gone (W1: 50 windows).

27 - On the ground-truthed clip, baseline F1@0.5 was ****0.000**** (mIoU 0.03); the best local config reaches ****0.296**** (mIoU 0.50). That is the real, measured win: from "unusable" to "detects most rallies, boundaries still loose."

28 - ****Detection recall is genuinely good:**** at ±0.25 tolerance, local finds ~86% of what Gemini finds on the ground-truthed clip. The remaining problem is **where the boundaries land**, not **whether rallies are found**.

29

30 ---

31

32 ## 3. What broke / new problems

33

34 **** (a) The over-segmentation pendulum is real and global.**** No single global setting hit ~1.0×: baseline 0.29× ? W1 1.52× ? W1+W2 1.35×. The cap swung us straight past parity into over-splitting.

35

36 **** (b) Per-venue variation defeats any single global config.**** Under **identical** W1 knobs, per-clip count ratio ranges ****0.22× ? 2.00×**** and mean window length ****7.8s ? 86.9s****:

37

38 | Clip | Gemini | W1 | ratio | mean window |

39 |---|---|---|---|---|

40 | GX010128 | 39 | 78 | 2.00× | 7.8s |

41 | GX010129 | 27 | 50 | 1.85× | 9.3s |

42 | GX020125 | 11 | 18 | 1.64× | 14.1s |

43 | mahadevpura-0 | 40 | 62 | 1.55× | 10.4s |

44 | mahadevpura-1 | 9 | 2 | 0.22× | ****86.9s**** (blob) |

45 | mahadevpura-2 | 39 | 40 | ****1.03×**** | 13.8s |

46

47 Tightening to fix GX over-split pushes mahadevpura-0/-1 further under; loosening to merge the blob worsens GX. The root cause is mechanical: the active-frame rule keys on ****"shuttle moving,"** not **"rally"** ? a window ends only when the shuttle goes invisible or drops below ``velocity_thresh`` (``tracknet_runner.py:305``, `end = last in-flight frame; `:285``, ``prev`` resets on any invisible frame; ``:292``, active requires `velocity > 0.02`). So inter-clip differences in **how the shuttle behaves between points** ? not rally structure ? drive the result. ****Honest caveat:**** this correlates with clip content/activity structure, ****not** cleanly with venue ? mahadevpura-0 over-splits like the GX clips (1.55×), so a "GoPro vs court" venue story is **not** supported at n=3/venue.

48

49 **** (c) The continuously-active blob (mahadevpura-1) is a distinct, unsolved failure mode.**** The clip has no inactivity gap ? ``merge_gap`` to split on, so ~174s coalesces. W1's cap **did** fire once (split into ~101.9s + ~71.9s) but both halves stay ~678× over the 12s cap ? there is no sub-gap to recurse on. Result: n=2, 0 matched, mIoU 0.0. The cap is ****under-effective, not a no-op**** here; W2 is a true no-op (W1+W2 byte-identical). This single clip's mIoU 0.0 must not be averaged into the headline.

50

51 **(d) W2 (net-crossings gate) buys count by spending recall, and only fires on one venue.** W1 ? W1+W2 dropped aggregate count 250 ? 222 (?28), but **~all of it is mahadevpura-0** (62?43, ?19, with **0?5 new gemini-only misses**); GX clips moved ?3 each, the blob 0. mIoU stayed flat at 0.39 and **6 net-new gemini-only misses** appeared. Golden lift was only **+0.019 (n.s.)** vs W1's +0.139. It violates do-no-harm for no ground-truth-backed gain.

52

53 **(e) Count parity is a deceptive proxy.** mahadevpura-2 lands at 1.03x ? near-perfect count ? yet matches only **12/39** with mean_iou 0.420 and F1@0.5 0.150 (W1+W2). Right count, wrong boundaries.

54

55 **(f) tIoU@0.5 over-penalizes and flips rankings.** For a 5.7s rally, tIoU 0.5 tolerates only ~±1.9s, and the ~1?1.5s service window nearly exhausts that budget. The hard-cap is golden-neutral at n=6 (+0.006 n.s.) but becomes *best* at n=7 once the continuously-active clip enters ? a metric artifact, not a real flip in quality.

56

57 ---

58

59 ## 4. The named next lever ? a rally-END signal (not a generic boundary head)

60

61 **Lever: a sustained-inactivity rally-END rule (See et al. style) in `trajectory\_to\_action\_windows`.**

62

63 **Why this specifically, and why not a generic boundary refiner:** the ground-truth boundary errors decompose cleanly ?

64

65 - **rally-START is already solved:** local |?start| median **1.74s ? Gemini 1.46s**.

66 - **rally-END is the entire gap:** local |?end| median **3.12s ? Gemini 1.19s**.

67

68 A generic "tighten both boundaries" head spends half its effort on START, which is already at reference parity ? wasted ROI and risk of regressing a good edge. The asymmetry says: **end the window at the last *sustained-active* frame, not at the last in-flight frame carried to the cap.**

69

70 Concretely: maintain a trailing window T (~1.5?2s); terminate the active group once ?80% of its frames are no-motion (velocity ? `velocity\_thresh` OR shuttle invisible), setting end at the *start* of that inactivity run rather than `group[-1][0]/fps` (:305). Critically, **decouple the END decision from single-frame visibility** ? stop treating one invisible frame as a hard trajectory break (:285) for the inactivity counter; count short occlusion bursts as "no-motion" contributions so brief end-of-rally occlusion doesn't cut early and a slow walk-back doesn't extend the window. This should tighten END on *every* venue without touching count on the well-behaved clips.

71

72 **Honest caveat on the END diagnosis:** the start?end-parity decomposition rests on the n=40 mahadevpura-2 ground-truth numbers above. An earlier, more aggressive framing (a cited `output/local\_vs\_gemini\_6.json` with per-matched-pair start/end stats across all 6 clips) **does not exist in this worktree** and must not be cited as if it does. The single-clip GT we *do* have is consistent and points the same direction, but the END-dominance claim is currently **n=1 ground-truth** ? confirm on GX010128 before treating it as settled.

73

74 ---

75

76 ## 5. Prioritized roadmap

77

78 | # | Lever | Effort | Expected payoff | Confidence | Depends on |

79 |---|---|---|---|---|---|

80 | **R1** | **Ship the smallest safe milestone as ONE flip: W1 cap=12 ON + W1.5 hard-cap (midpoint-chop when no gap) ON, no W2.** Best-measured local config; hard-cap fires only when W1 finds no gap, so it's a no-op on the 4 gap-segmented clips. Assert never-empty (?1 window, `min\_window`=2.0s) as a tested invariant. | S | F1@0.5 0.150?0.296, F1@0.25 0.500?0.722, mIoU 0.31?0.50 on the GT clip; rescues the blob from the degenerate 1?2 window outcome | High (measured on GT clip) | ? |

81 | **R2** | **New primary eval metric: ±2s tolerance-match F1 + boundary-error-in-seconds, with over-seg guard.** Asymmetric Hungarian one-to-one match (?_start=2.0s, ?_end=1.0?1.5s); report P@?, R@?, F1@?; split/merge counts per video; signed+abs ?start/?end

median+IQR. Demote tIoU/mIoU to secondary diagnostic. | S?M | Stops the count-parity trap and the tIoU over-penalty; makes over-seg cost precision directly; tells us start is solved / end is open | High | ? |

82 | ****R3**** | ****Re-scope the cap to true rally length**** ? test `max\_window` in the ****6?8s**** band (+ separate end-trim) instead of 12s; the hard-cap lift already signals 12s is coarse. Ablate cap-only vs hard-cap-chop on the GT clip, holding velocity_thresh/merge_gap fixed. | S?M | Less END over-extension and less over-counting on short-rally venues | Medium (12s coarseness measured; 6?8s untested) | R2 (need the metric to score it) |

83 | ****R4**** | ****Sustained-inactivity END rule**** (§4): trailing-window 80%-no-motion termination; decouple END from single-frame visibility. Config-gated, default-OFF. | M | Directly attacks the |?end| 3.12s?target ?~1.2s (Gemini parity); helps every venue, count-neutral on clean clips | Medium-High (root-cause-aligned; needs n>1 GT) | R2, R3 |

84 | ****R5**** | ****Blob splitter for the gap-less case**** (mahadevpura-1): when the inactivity rule never fires within an over-cap group, fall back to W1.5 midpoint-chop ****and log it as a forced cut**** flagging the clip as needing rally-STATE signals (net-crossings / two-sided motion), not a bigger cap. Track as its own eval stratum. | M | Removes the mIoU 0.0 / 0-match pathology on continuously-active clips; prevents one clip swinging the mean | Medium | R1, R4 |

85 | ****R6**** | ****Expand + stratify golden set to ~16 by the weekend****, over-sampling continuously-active clips and each venue. Re-fit ? against a genuine inter-annotator study (2nd labeler re-labels 2?3 Roora videos; set ? to 90th-pct pairwise human \{|?|\}). Re-run LOVO + sign-test before locking any default. | M | Stops single-clip rank flips (hard-cap n.s.@6 ? best@7); makes ? defensible | ? | R2 |

86 | ****R7**** | ****Keep W2 (net-crossings) OPT-IN / default-OFF****, retained as a flag-gated scorer/post-filter so it keeps accruing measurements (weak-signals-retained directive). Do ***not*** promote. | S | None now; future signal for the corpus | High (the harm is measured: 6 net-new misses, +0.019 n.s.) | ? |

87 | ****R8**** | ****Do NOT build a serve/START classifier**** (See et al. M1) until END lands. START is at Gemini parity; marginal ROI is on END. Revisit only after END closes. | ? | Avoids wasted effort | High | R4 outcome |

88

89 ****Default-ON config is GLOBAL, not per-venue.**** The hard-cap's fire-only-when-no-gap behavior is already content-adaptive, so it covers the continuously-active case without a brittle per-venue switch fit on n=1. Revisit per-venue presets only when the corpus has meaningful per-venue n (well beyond ~16).

90

91 ---

92

93 **## 6. Eval methodology change (gate the roadmap on this)**

94

95 Replace headline tIoU-F1 with a ****task-faithful tolerance metric****:

96

97 1. ****Tolerance-match F1 (PRIMARY).**** A prediction matches a reference iff `|?start| ? ?\_start AND |?end| ? ?\_end`, asymmetric: ****?_start = 2.0s**** (~2x the service-window/inter-annotator start noise; Gemini ?start median 1.46s), ****?_end = 1.0?1.5s**** (rally-end is a sharp event; Gemini ?end median 1.19s, and it's exactly where local fails at 3.12s). Freeze these as a documented decision in `docs/GOLDEN\_REGRESSION\_FIXTURES.md` with the IoU?time derivation `? = D.(1??)/(1+?)` recorded so the choice is auditable, until R6's inter-annotator study refines them.

98 2. ****Over-segmentation guard (MANDATORY).**** Strict one-to-one matching via ****Hungarian** global assignment, never greedy.** Report ****P@? = matched/n_pred**** alongside ****R@? = matched/n_ref****. This makes the 222/165 = 1.35 surplus cost precision directly (mahadevpura-2: 40 preds vs 40 refs, 12 matched ? P=0.30, R=0.30 ? over-seg fully visible; widening ? can't hide it).

99 3. ****Fragmentation/merge pair (SECOND guard, no-regression-gated per video).**** split_count (predictions covering one reference ? catches W1 over-seg) and merge_count (references under one prediction ? catches the baseline mega-window pathology). ****A promotion may not increase either guard on ANY video even if mean F1 rises**** ? this blocks the mahadevpura-0 trap (count parity improved 1.55?1.075 while gemini_only misses went 0?5).

100 4. ****Boundary error as TWO separate distributions**** (median + IQR of signed and absolute ?start, ?end over matched pairs), never folded into one IoU. This is the metric that says start is solved and end is the open problem, and directs effort to the velocity-decay / net-crossing tail.

101 5. ****tIoU/mIoU stays as a SECONDARY diagnostic**** ? single-number trend line, ties to the paper ablation backbone ? but is ****not the gate.****

102

```

103     **Promotion (trustworthiness) rule at n=6??:** a delta is promotable only if *(1)** the
paired per-video sign-test on tolerance-F1 is a near-unanimous sweep ? **6/0 at n=6 (p=0.031)**
or **7/0 at n=7 (p=0.016)**; 5/1 or 6/1 is NOT significant (p=0.219 / 0.125); **AND (2)** over-
seg and merge guards do not regress on any video; **AND (3)** the effect size is non-negligible
(mean delta + bootstrap CI as an honesty band ? expect it to span 0 at this n; use it to
separate baseline?W1 +0.139 "real" from W1?W1+W2 +0.019 "directionally-clean-but-negligible,"
not as a pass/fail gate). **Stratify, don't blend:** report mahadevpura-1 as its own stratum.
104
105     ---
106
107     ## 7. Honest caveats
108
109     - **n=6 clips, exactly ONE with ground truth (mahadevpura-2).** Every tight-boundary
number (F1@0.5, |?start|/?end|, the start-solved/end-open decomposition) rests on that single
owner-labeled clip. **GX010128 is the next ground-truth target** to confirm the END-dominance
diagnosis on a second, denser clip before locking R4.
110     - **Gemini is a strong REFERENCE, not ground truth.** The other 5 clips are scored
against Gemini; "1.52x of Gemini" measures agreement with Gemini, not correctness. Gemini stays
strictly an **inference-time reference, never a training-data source** (licensing guardrail).
111     - **Single-clip effects are load-bearing and fragile.** W2's count reduction is
~entirely mahadevpura-0; the hard-cap's "best at n=7" status is driven by the continuously-
active clip; the blob's mIoU 0.0 can swing a 6-clip mean. This is exactly why R6 (stratify +
expand to ~16) gates locking any default.
112     - **Do not over-claim "at par."** Local is **~86% of Gemini on rally DETECTION (F1@0.25
0.722 vs 0.835)** but only **~53% on tight boundaries (F1@0.5 0.296 vs 0.557)**. START is at
parity; the headline number people will quote should be the honest split, not a single rounded
figure.
113     - **Refuted claims explicitly NOT relied on:** there is no
`output/local_vs_gemini_6.json` in this worktree, so any per-matched-pair start/end stats
*across all 6 clips* are unsubstantiated; the only local DB present (`output/val.db`) covers
GX010128, GX010129, mahadevpura-0 (not mahadevpura-2). The "cap=12 is the proximate cause of END
over-extension" and "GX010128 chops single true rallies" claims are **not** supported by the
present data (GX010128's inflation is dominated by 34 local_only false-positive windows, splits
metric only 4; its W1 windows average 7.76s, *longer* than Gemini's 5.85s). Treat cap=12 as *one
contributor* alongside detector recall and the 2s merge_gap, to be isolated by R3's ablation ?
not as the established root cause.
114     - **All proposed flags ship default-OFF**, promoted only on the $6 rule. The only thing
recommended for default-ON is the R1 milestone (W1 + hard-cap, no W2), and even that is **gated
on a re-run at the ~16-clip corpus** since the hard-cap was golden-neutral at n=6 and only
became best at n=7.
115
116     ---
117
118     *Files referenced (absolute):*
119     `C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\GOLDEN_REGRESSION_FIXTURES.md` (tracked doc to fold
this into) · `...\docs\RALLY_QUALITY_RESEARCH.md` $2 (root-cause backbone) · code anchors in
`backend\pipeline\segmenters\...\tracknet_runner.py` lines 285, 292, 305, 358?364.

```

11. user (2026-06-19T15:46:49.267Z)

```

1     # Post-#166 Work Items, Milestones, Risk Assessment, and Sequencing Plan
2
3     **Date:** 2026-06-14 (post-merge of #166)
4     **Status:** Plan for owner review first. Execution only after explicit owner approval
("review first and then execute"; "I want a plan detail as to what are the next items planned,
please execute only after I explicitly say so").
5     **Purpose:** Per owner directive after #166 merges: stop the prior babysit loop, deliver
this as a new project MD PR for review (with full risk assessment like the T0-T5 in the archived
audit, sequencing rationale, and broken into milestones), then (only after review + explicit go)
execute one PR per slice off `master`.
6
7     **Key outcomes from #166 (merged c558372):** Hardening loop (T1-T5 + finale #165)
COMPLETE and archived. NEXT_STEPS reframed with explicit P3 start signal (harness LOVO litmus
for person features) + full prioritized candidate list. Gate A wired (opt-in, default-OFF safety

```

```

per served regression litmus). Loop scheduler (019ec6d07da5) deleted; no replacement created.
Handoff and archived audit updated terminal. Master also includes #167 (packaging).
8
9     **Sources (live, read these for details):**
10     - `docs/NEXT_STEPS.md` (prioritized 1-6 + rally track pick-up order)
11     - `docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3 learned fusion + P4 audio)
12     - `docs/EXPERIMENT_HARNESS.md` (A/B, shadow, LOVO-honest `compare`, promotion gate, flag
default-OFF, rollback=config flip)
13     - `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (M0.1?M0.4 Phase-0, event-index contract,
TAL head first, ms-swift + Qwen3-VL, guardrails)
14     - `docs/BACKLOG.md` (rename ?, cost benchmark as largest open risk, detector follow-ups,
golden subproject, etc.)
15     - `docs/DEFERRED_HARDENING_PLAN.md` (remaining: train/eval separation guardrails)
16     - Archived: `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (T0-T5
risk style + Living Log precedent), `docs/archives/HARDENING_LOOP_HANDOFF-
COMPLETED-2026-06-14.md`, CLAUDE.md, `docs/HOW_RALLY_DETECTION_WORKS.md`
17
18     **Process guardrails (non-negotiable, carried forward):** ONE PR per slice/item off
`master`; no stacking. Branch, implement (flag-gated default-OFF for any quality/cue changes),
run available tests + post-change guards (syntax/ast here; owner re-runs full `run_all_tests.py
--cov...` + real-video + golden LOVO/ablation on hardware), commit descriptively, push, create
PR, babysit (poll/fix/reply via MCP + handoff pattern) until owner merge observed ? then
advance. Never self-merge. For quality experiments: use harness `compare` (LOVO + honest
stats/CIs/sign test), `run_regression.py` (exit 0/1/3 gate), record to
`eval_baselines/experiment_leaderboard.json`. Promote only on measured lift + no regression.
Container limits: use `python -c "import ast; ..."` + note "All green where executable. Owner
re-runs full on hardware."
19
20     ### Transition Record (from #166 merge + owner stop-loop directive)
21     #166 merged (c558372, LGTM 4702279497). Per explicit owner request after #166: "stop
this loop and send a PR for these work items as a new project md file so we can review first and
then execute this and do assess the risk so we can sequence them well. Would be good to break
them into milestones." Scheduler 019ec6d07da5 deleted; no replacement. Handoff trimmed to
pointer (per #165 review intent); detailed record here + in plan. All T1-T5 + #165 + #166
transition complete. Clean slate for review of this plan before any further execution.
22
23     ---
24
25     ## Work Items Catalog (with risk per T0-T5 precedent)
26
27     Risk levels (from archived audit §3 style):
28     - **LOW**: docs-only, config, pure additive tests, no behavior impact.
29     - **LOW-MEDIUM**: new/expanded positive tests or thin pure-logic glue.
30     - **MEDIUM**: new runtime paths (even if gated/flag default-OFF), new MIT/Apache deps,
feature persistence, small harness increments, real-video validation.
31     - **MEDIUM-HIGH**: DB migrations + new data contracts (event store), train/eval
guardrails (correctness of moat), compliance changes to training sources.
32     - **HIGH**: anything touching "paid-LLM as training data" guardrail, unquantified large
costs (Gen-2 VRAM/$), production default flips without lift proof, licensing/C3 blockers,
multisport generalization hypothesis.
33
34     1. **Grow the golden corpus (NEXT_STEPS #1 ? highest leverage)**
35     Priority: multi-court badminton (target distro + ship target), then ?3 matches per
new sport (TT first; pickleball labels exist but ball detector blocker). Adult sessions only
(minors-exclusion). Per video: label (Roora v8 + guidelines) ? curate import-local (Proprietary
license) ? re-run harness.
36     **Risk: LOW** (mostly owner/data-collection + collector cross-repo; import is
existing). Small-N statistics power improves with n=10+ (sign test reachable).
37     **Verification:** manifest growth, re-harness LOVO/sign test on held-out folds, owner
real footage.
38
39     2. **Ratify ship-gate target (NEXT_STEPS #2 + BACKLOG + OWNED M0.4)**
40     Floor = beat heuristic LOVO ~0.480 (necessary). Target informed by data: multi-court
wrapper 0.66 (the number that makes owned worth serving); clean 0.93 is ceiling, not target.
Shape suggestion: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs

```

heuristic".

41 ****Risk: LOW**** (docs/decision only).

42

43 3. ****Rally quality P3 (learned person fusion) + P4 audio (FUSION \$10 + EXPERIMENT_HARNESS P2/P3)****

44 P3: integrate person features (from the P2 FusionSegmenter + `fusion\_features`) into the harness `compare` for LOVO litmus on golden videos; report ?F1 + honest stats/CIs/sign test. Eager/windowed compute is in place from P2. If lift: optional served gating (flag default-OFF).

45 ****Investigation note (updated per latest review):**** The LOVO compare driver (`fusion\_compare.py`), golden extractor (`fusion\_golden.py`), ablation runner (`ablation.py`), and related tests (`test\_fusion\_compare.py` etc.) ****are present**** in the tree (verified via `git ls-files` + ls on backend/eval/ and tests/). Person feature computation lives in `backend/pipeline/segmenters/fusion\_hybrid.py::\_person\_features` (and `backend/pipeline/detectors/fusion\_features.py`). The remaining in-container P3 work for M0 is wiring the person features into the `compare` path (leveraging the existing harness seam for extra feature columns in the LogisticRegression) + running the litmus on the golden substrate. GPU extraction over the full corpus (to produce the real feature JSONs) remains the hardware-gated step. This makes the M0 placement of item 3 even more appropriate as a quick in-container win (the measurement machinery is shipped; the integration + run is the task).

46 P4: wire audio (`hit\_detector`) + re-eval (prior E1 found weak; cheap re-probe). Full learned promotion only on measured lift.

47 ****Risk: LOW-MEDIUM** for pure in-container harness litmus glue + test; **MEDIUM overall**** (including gated served path + owner real LOVO).

48 ****Verification:**** harness `compare` (deterministic on synthetic + golden), no-regression invariant, owner real-video + ablation; `run\_regression.py` gate.

49

50 4. ****M0.4 Gen-0 TAL harness (OWNED M0.4 + NEXT_STEPS #3)****

51 Swap Gen-0 LogReg for ActionFormer (or ASFormer; MIT, lightweight, minutes on 2070S) inside existing harness (`training/gen0/harness.py` reuse `metrics.match\_intervals`). Per-video threshold calibration (LOVO-vs-oracle gap visible per fold). One-command train?eval?ship-gate. Define ship target up-front (see item 2).

52 ****Risk: MEDIUM**** (new MIT dep in training, code change to harness, but local \$0, reuses featurizer/metrics, no production path until gate passed).

53

54 5. ****Owned Phase-0 foundation ? M0.1 corpus spine + event-index contract (OWNED M0.1)****

55 Canonical JSONL per rally (video_id MD5, rally_ordinal, sport, start/end, shot_count, ending_reason/fault_tags, label_type, source, consent/training_opt_in, split=BY MATCH, trajectory_ref, ...). Wire to durable per-video ****event store**** (the conversational-QA seam: "extract once, query many"). DB delta: extend `play\_segments`, new `highlights`/events table + `PRAGMA user\_version` migration. Reserve fault/event taxonomy now (service_fault etc.). No Gemini-sourced rows in training views.

56 ****Risk: MEDIUM-HIGH**** (real DB migration + new first-class data contract affecting all future conversational features; must be correct early).

57

58 6a. ****M0.3 Gemini-silver purge from training paths (OWNED M0.3)**** ? retarget `distill\_local`/`calibrate\_local` etc. to human + open-teacher labels only (Gemini eval/fallback forever). Core moat-correctness code change.

59 ****Risk: HIGH**** (guardrail "never train on paid-LLM").

60

61 6b. ****Train/eval separation guardrails ? pure-logic core (DEFERRED #13 + BACKLOG)**** ? `backend/eval/splits.py::assert\_disjoint(train\_ids, eval\_ids)` + unit test + raise-on-overlap. Split unit = match. (Defer collector manifest `split` field wiring to later cross-repo step.) In-container, zero behavior risk.

62 ****Risk: LOW-MEDIUM**** (pure logic + tests; unblocks clean baselines).

63

64 6c. ****WASB C3 + related commercial blockers (tracking only)**** ? first-class tracked item (own-weights or clean detector; multiplies per sport). No code change in this repo until resolved upstream.

65 ****Risk: (commercial blocker, not code risk).****

66

67 7. ****Multisport thread + clean detectors (NEXT_STEPS #4 + OWNED + BACKLOG)****

68 Ingest Roora pickleball/TT CSVs (`import-local --normalize`). Merged-prototype LOVO across badminton+TT once TT has ?3 matches. Identify clean MIT/Apache/BSD ball/shuttle detector per sport (WASB C3 is unresolved blocker for some paths; TT-on-WASB serving gated). Single-model-multisport-temporal-generalization is unproven hypothesis (validate on 2nd-sport corpus).

69 ****Risk: MEDIUM**** (data + per-sport detector hunt; C3 is commercial blocker).

70

71 8. ****Housekeeping ? rename repo off "badminton" (BACKLOG ? + OWNED)****

72 Owner-emphatic "ASAP" (project is multisport). Touches remote name, README/CLAUDE

73 ****Risk: LOW-MEDIUM**** (docs + references + coordination; breaking for external

74 references until done).

75 9. ****PMF validation (NEXT_STEPS #5)****

76 C13 academy interviews (add the two questions on multi-court + pay-per-match). Camera

77 pilot at 2-3 warmest academies (consent at capture; demo via Gemini or human-polish reel). Field

78 kit complete (flyer + playbook + answer sheet + pre-registered G/A/R signals +

79 PROCEED/PIVOT/FALSIFIED thresholds). Owner-led (hired associate or direct).

80 ****Risk: LOW**** (mostly non-code / owner field work; cheap vs engineering month).

81 10a. ****In-container chores (low-regret, executable now)**** ? WSL/Gemini remote-file GC +

82 retention policy note; tests reorganization (per owner-specified grouping metric from prior

83 reviews); other small docs/code hygiene from BACKLOG.

84 ****Risk: LOW****.

85 10b. ****Hardware / open-risk items (triggered, owner-gated)**** ? Quantify cost-to-Gen-2

86 (the single largest open risk: real VRAM/token/\$ on A100/L40S before any owned serving);

87 ByteTrack deprecation + real-footage person detector validation; golden subproject follow-ups

88 that require hardware/corpus.

89 ****Risk: MEDIUM-HIGH**** (unquantified cost is load-bearing; real video needed).

90 ---

91 ## Risk Assessment Summary Table (unbundled per review)

92

ID	Item	Risk	Rationale / Deps	Verification
1	Grow golden corpus	LOW	Data effort + existing curate; consent guard	Manifest + LOVO/sign test improvement
2	Ship-gate target	LOW	Docs/decision; unblocks harness error	Explicit ratified number in plan
3	P3/P4 fusion litmus + harness (in-container core + gated)	LOW-MED (core) / MEDIUM (full)	General harness present; specific person-feature LOVO driver + GPU extraction are the work. Gated promotion.	harness `compare` + honest stats + regression gate + owner ablation/real LOVO
4	M0.4 ActionFormer harness	MEDIUM	MIT dep + harness change (local \$0)	One-command + per-fold cal + ship-gate
5	M0.1 event-index contract + spine	MEDIUM-HIGH	DB migration + new contract	Round-trip, split-by-match, no Gemini train rows
6a	Gemini-silver purge (training paths)	HIGH	Core moat guardrail correctness	Training entrypoints retargeted; no Gemini-silver in train views
6b	Train/eval guard ? pure logic core	LOW-MEDIUM	`splits.py` assert + test (in-container)	Unit test + raise-on-overlap; defers collector wiring
6c	C3 / commercial blockers	(tracking)	Multiplies per sport; gates ship	Tracked in BACKLOG/OWNED; no code until resolved
7	Multisport + detectors	MEDIUM	Data + per-sport detector hunt (C3 multiplies)	?3 TT matches + cross-LOVO prototype
8	Rename	LOW-MEDIUM	References + remote + framing	Clean grep + owner remote action
9	PMF field	LOW	Owner field work (kit ready)	Interviews logged + signals thresholds
10a	In-container chores (GC, tests-reorg, etc.)	LOW	Small hygiene / docs	Grep-clean, tests re-org per owner metric
10b	Cost-to-Gen-2 benchmark + real validations	MEDIUM-HIGH	Unquantified \$ (largest open risk); needs A100/L40S + real footage	Benchmark numbers + reports before any Gen-2 work

104

105 ****Overall sequencing risk mitigation (revised per course-correction):**** Front-load M0

in-container quick wins (high-value engineering with zero external blockers). Owner-led work

(corpus growth as #1 leverage, PMF, rename) runs in parallel. Heavy/gated work (DB migration,

purges, cost benchmark) only after corpus + ratified ship target + quick wins are in place. All quality items flag default-OFF, promote only on measured lift via harness + regression gate + owner real-video. Owner explicit "proceed with X" required before any implementation.

106

107 ---

108

109 ## Milestones (revised per review course-correction ? front-load in-container quick wins)

110

111 **Milestone 0: Quick Wins (in-container engineering; unblocked by corpus, hardware, or additional owner gates ? do these now)**

112 These are the low-hanging, high-value items that can be executed in the dev container with the existing harness + pure logic. Per the rebuttal: front-load these so the cheapest engineering wins aren't buried behind heavy owned-model / corpus work.

113

114 1. **Ratify the ship-gate target (item 2).** Pure decision/docs. Clears the live hard-error in `training/gen0/harness.py` on "undefined F1@tIoU target". Numbers already exist in the data (floor ~0.480 heuristic, multi-court reference 0.66, clean ceiling 0.93). Highest-value quick win; also unblocks M0.4 gate and regression baselines.

115 **Risk: LOW**. In-container.

116

117 2. **Pure-logic core of train/eval guard (item 6b).** Implement `backend/eval/splits.py::assert\_disjoint(train\_ids, eval\_ids)` + raise-on-overlap unit test (split by match). The spec is in BACKLOG/DEFERRED. Fully testable here; defer collector manifest wiring.

118 **Risk: LOW-MEDIUM**. In-container.

119

120 3. **Run / land the P3 fusion LOVO litmus (item 3 core).** Integrate person features into harness `compare` (or dedicated driver) and execute on existing/synthetic features + the 6 golden (LOVO + honest stats). Note: general experiment harness + P2 fusion segmenter/person persistence are present; the specific P3 litmus driver over fusion features is the implementation task (see investigation in catalog item 3). GPU extraction over full corpus is the hardware-gated part.

121 **Risk: LOW-MEDIUM** (in-container glue + tests).

122

123 4. **In-container chores from item 10a.** WSL/Gemini GC + retention policy; tests re-org (owner grouping metric); other small hygiene. Separate from hardware cost benchmark.

124 **Risk: LOW**.

125

126 *Parallel / owner-led (not primary agent engineering focus):* Grow the golden corpus (your #1 leverage item), PMF field visits (using the existing kit), repo rename. These run alongside M0 without blocking the quick engineering wins.

127

128 **Milestone 1 ? Rally Quality P3 completion + ActionFormer (after corpus growth + ratified ship target)**

129 - Complete P3 (GPU litmus on real corpus + any served wiring on proven lift) + P4 audio re-probe.

130 - 4. M0.4 ActionFormer harness (one-command, per-video cal, integrated with the now-ratified ship-gate).

131 *Rationale:* P3/ActionFormer need the corpus for meaningful LOVO/significance and a clear target to gate against. After M0 quick wins + owner corpus work.

132

133 **Milestone 2 ? Owned Phase-0 heavy foundation (explicitly gated; last for agent engineering)**

134 - 5. M0.1 event-index contract + corpus spine (DB migration).

135 - 6a/6c. Gemini-silver purge (moat code) + C3 tracking / commercial resolution.

136 - 10b. Cost-to-Gen-2 benchmark + real-footage validations (the load-bearing open risks).

137 *Rationale:* These have the highest blast radius, largest unquantified risks, or external blockers. Do after the quick wins have landed value and the corpus/target are solid. DB contract can be designed in parallel but implemented late.

138

139 **Milestone 3 ? Multisport, Rename, PMF (owner-led + triggered)**

140 - 7?9 and remaining: multisport TT prototype, rename, PMF execution, etc.

141 These are largely owner/field or data-triggered.

142

```

143     **Later / explicitly gated (do not start without owner go after prior milestones):**
Phase-1 ship decision, Phase-2 Gen-1/Gen-2 (after cost benchmark + quality bar cleared), etc.
144
145     This re-sequencing directly implements the course-correction: quick in-container wins
first (M0), owner-led in parallel, heavy/gated work after data + targets, comprehensive catalog
+ risk table preserved for reference.
146
147     ---
148
149     ## How to Use This Plan (after this PR is reviewed + merged)
150
151     1. Owner reviews this MD (risks, sequencing, milestones).
152     2. Owner explicitly says (comment or new issue): "LGTM, proceed with Milestone 1" (or
specific item, or adjusted order).
153     3. Agent: re-read handoff (or new lightweight handoff if needed), branch off master for
the item, implement per discipline (gated, tests, harness where applicable), pre/post baselines,
PR, babysit until observed merge, record, advance.
154     4. Update this plan MD + NEXT_STEPS + living logs on each step (append-only).
155     5. Repeat per milestone/item. Revisit this MD for risk/sequencing if priorities shift.
156
157     This keeps the "review first" + "explicitly say so" contract while giving a sequenced,
risk-assessed path forward.
158
159     ---
160
161     ### 2026-06-14 ? This PR: M0 "backend/eval/splits.py train/eval guard" (splits
assert_disjoint + tests)
162     - Branched off master (post #169 merge f000324), one PR per item per plan + handoff
rules.
163     - Created `backend/eval/splits.py`: `load_split(manifest)` (list or path -> (train_ids,
eval_ids) sets keyed by match id, not MD5; supports 'split' or defaults to train) +
`assert_disjoint(train_ids, eval_ids)` that raises ValueError on overlap with message including
"This would have caught train/eval leakage."
164     - Created `tests/test_splits.py`: 4 tests (load+disjoint happy, overlap raises with
leakage comment, disjoint passes, path load). Explicit "would have caught leakage" comment.
165     - Deferred per query: cross-repo collector manifest `split` field wiring + integration
into distill_local/training/run_regression (later PR).
166     - Pre-change baseline (before edits): syntax on touched, no prior splits.py.
167     - Full gate (container: no pytest/torch etc.):
168     - `python run_all_tests.py --cov=backend --cov=training --cov-fail-under=70`: printed
obsreport note; "No module named pytest" (exit non-0 as expected). "All green where executable
in this env. Owner must re-run full baseline + real-video/golden LOVO on capable machine (with
pytest/cov) before LGTM."
169     - `ruff check .`: (executed; results in session log - clean or fixed if any).
170     - `mypy backend training`: (executed; results in session log).
171     - Post-edit smoke: `python -c "from backend.eval.splits import load_split,
assert_disjoint; import ast; ast.parse(open('tests/test_splits.py').read()); print('import+ast
OK')"` ? passed.
172     - This PR babysat (MCP poll for comments/reviews/fix if needed using add_issue_comment
etc.) until owner merge observed, then advance to next M0 item (P3 litmus).
173     - Living docs: this append-only entry in plan MD (NEXT_STEPS.md may get note in hygiene
slice or later).
174     - Per plan M0 + query spec + handoff rules (re-read M0 + handoff before item; branch
master; full gate + post results; no self-merge; update docs append-only).
175
176     ### 2026-06-14 ? Review round for #171 (splits guard)
177     - Owner review (detailed): must-fix 1-4 to make guard fail loud never silent (1. non-
list manifest vacuous pass ? add list check+raise; 2. bad id-res or chain ? explicit not-None
non-empty, raise if no real match key, avoid name/video_id; 3. missing tests for core leakage
paths (same match_id diff video, leaked manifest end-to-end via load+assert, non-list/empty id)
with comments; 4. non-strict split label ? whitelist train/eval raise else). Should: pathlib,
utf-8-sig. Gate green but fixes needed.
178     - Addressed exactly (minimal, on current branch):
179     - splits.py: list validation+raise after load (for json/direct); explicit not-
None+non-empty mid check with raise if no real match key (match_id then id); no fallback to

```

```

name/video_id; strict sp in ("train","eval") else raise; utf-8-sig; os.PathLike in
sig+isinstance (pathlib support).
180     - test_splits.py: added tests for non-list raises, same match_id diff video_id flags,
end-to-end leaked via load_split+assert, falsy/empty id raises, bad split label raises; all with
"this would have caught train/eval leakage" comments. Updated existing for coverage.
181     - plan MD: this append-only review round entry.
182     - Pre-edit gate (syntax): OK. ruff/mypy not in container env (as always); "All green
where executable. Owner re-runs full on hardware."
183     - Committed as "address review: ...", pushed. add_issue_comment with "Addressed in
<sha>... Ready for re-review."
184     - This PR babysat until merge observed (per rules), then advance (after 170 too).
185
186     ### 2026-06-14 ? Re-review round for #171 (splits guard)
187     - Owner re-review (post-address): CI red / mergeable_state=unstable. Blocker traced to
`test_load_split_non_list_raises`: direct-dict case `splits.load_split({"foo": "bar"})` did not
raise (returned empty sets) while path-based non-list json did raise. Root cause in the else
(direct manifest) branch: `data = list(manifest); if not isinstance(data, list)` ? `list(dict)`
yields list-of-keys (truthy list), so guard bypassed; loop skips non-dict keys ? vacuous clean.
(Path json branch caught before listify.)
188     - Addressed exactly (minimal fix per review suggestion):
189     - splits.py: in non-path else, add `if isinstance(manifest, dict): raise
ValueError(f"manifest must be a list of entries, got {type(manifest)}")` *before* `data =
list(manifest)`. Matches error text used by test + json branch. Keeps all prior loud-fail
hardening.
190     - Verified: direct dict now raises correctly; path non-list still; happy paths +
strict + leakage tests still pass; no behavior change for valid manifests.
191     - Smoke gate (container limits): `python -c "from backend.eval.splits import ...;
import ast; ast.parse(...)"` + explicit repro of the dict case + happy/strict: all OK.
192     - plan MD: this append-only re-review entry (plus prior review round).
193     - Full gate (executable here): `python run_all_tests.py --cov...`: "No module named
pytest" (obsreport note) as expected; "All green where executable in this env. Owner must re-run
full baseline (662+ passed / 79.26% cov per prior review) + real-video/golden LOVO on capable
machine before LGTM." ruff/mypy unavailable in container (same as always).
194     - Committed as "address re-review for #171: ...", pushed. GH add_issue_comment
"Addressed in <sha>...".
195     - Per owner note in re-review: #170 is clean/green; merge #170 first (then rebase #171
on updated master to pick plan-MD changes from 170 + resolve any append conflict). Babysit
continues (both must be observed merged before advance per rules). No self-merge.
196     - Scheduler active for 2m poll (019ec71467c3); will re-read this plan MD on next tick.
197
198     ---
199
200     *Created as the requested new project MD after #166 merge. All per owner instructions,
CLAUDE.md guardrails (commercial-clean licensing, no paid-LLM training data, etc.), and loop
rules (no self-merge, one PR, observed-merge advance).*
201
202     **End of plan.** Review and give explicit go/no-go + any adjustments.

```

12. assistant (2026-06-19T15:46:52.879Z)

Now let me verify the key claims against the actual code. The central question across these docs is whether the "milestone" (cap=6 + R4 windowing) is still default-OFF/owner-gated, since PR #204 ("R1 milestone ? flip windowing defaults ON (cap=6 + R4)") appears in git history. Let me check the config defaults and that PR.

13. user (2026-06-19T15:46:55.129Z)

```

67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at
n=13 (#204)
---SHOW STAT---
commit 67b69279a4b97fb9d1283d07863f8a1721dc0dfe
Author: Avi <avi.dullu@gmail.com>
Date: Thu Jun 18 11:23:56 2026 +0530

```

feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13

(#204)

Flips the shipped local-windowing defaults to the R1 milestone (owner product-go granted):

```
max_window_duration    0.0  -> 6.0    (W1 bounded windowing, R3 cap)
window_inactivity_end   0.0  -> 1.5    (R4 rally-END inactivity trim)
inactivity_rally_thresh 0.08  -> 0.20
inactivity_min_density  0.34  -> 0.30
```

W2 net-crossings, W1.5 hard-cap, and R5 blob-fallback all stay default-OFF.

Re-verified on the grown n=13 golden corpus (the 2 new Boxhill clips GX010137/GX010141) ? the

milestone holds and is stronger than at n=11: baseline->milestone tolF1 ?+0.222

[+0.152,+0.286],

sign 12/0/1, p=0.0005; cap6 beats cap12 ?+0.110, sign 12/0/1; net over-seg guard PASS (mean merge_rate 0.651->0.161, zero per-video regressions). Both new clips improve. Independently reproduced by a 2nd harness (#214 lovo_resweep, matches to 3dp; LOVO picks cap=6 13/13 on

tIoU)

and survived a 5-lens adversarial workflow (every L00 fold clears the bar).

Ships with the mandatory Launch Report (docs/R1_MILESTONE_LAUNCH_REPORT.md ? both rulers + the

over-seg guard + honest caveats) and a QUALITY_ITERATIONS "R1 LAUNCHED" entry. Updates the 3 default-OFF assertion tests to the new default and the nightly tracked-configs docstrings

(its

"default" config = SERVED now equals "milestone"; its config-drift detector self-flags the

frozen

baseline STALE post-flip -> re-baseline once with --update-baseline).

Full suite 855 passed, coverage 80.72%; ruff + mypy clean.

Co-authored-by: Claude Opus 4.8 <noreply@anthropic.com>

```
backend/config/models.py          | 30 ++++++-----
docs/QUALITY_ITERATIONS.md        | 24 ++++++++
docs/R1_MILESTONE_LAUNCH_REPORT.md | 88 ++++++
scripts/nightly_regression.py      | 20 +++++--
tests/test_nightly_regression.py   | 14 +----
tests/test_windowing.py            | 29 ++++++-----
6 files changed, 169 insertions(+), 36 deletions(-)
```

14. user (2026-06-19T15:46:56.115Z)

backend\config\models.py:90: threshold is raised (`min\_crossings`/`min\_players` default 0 = OFF). The court mask

backend\config\models.py:105: min_crossings: int = Field(default=0, ge=0)

backend\config\models.py:136: # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip. Measured n=13

backend\config\models.py:137: # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below)

?+0.222, sign 12/0, p=0.0005; cap=6

backend\config\models.py:138: # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate 0.651?0.161). ***

backend\config\models.py:139: max_window_duration: float = 6.0

backend\config\models.py:143: # chopped at its midpoint until ? max_window_duration, making the cap a true upper bound.

backend\config\models.py:145: window_hard_cap: bool = False

backend\config\models.py:146: # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving slowly

backend\config\models.py:150: # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but significant

backend\config\models.py:151: # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, [?end] 4.96?3.31s (n=13). The W1 cap

backend\config\models.py:152: # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md "R1". ***

backend\config\models.py:157: # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group STILL longer

```

backend\config\models.py:158:    # than window_blob_factor * max_window_duration (no internal
gap, no rally-end signal ? e.g.
backend\config\models.py:159:    # mahadevpura-1's ~65s residual ? 11x a 6s cap) is midpoint-
chopped to ? the cap as a FORCED cut
backend\config\models.py:161:    # clip is surfaced as needing rally-STATE cues, not a bigger
cap. Differs from window_hard_cap
backend\config\models.py:162:    # (which chops BEFORE R4 and over-segments normal clips): blob-
fallback runs AFTER R4 and only
backend\config\models.py:165:    # Magnitude gate that keeps R5 surgical: only a group longer
than this * max_window_duration is a
backend\config\models.py:166:    # blob (a normal long rally ~1-2x the cap is left alone; the
mahadevpura-1 residual is ~11x).
backend\config\models.py:175:    # chunks of a high-bitrate (4K) source blow past the provider
upload cap
backend\config\models.py:274:    # Whole-file cap (default 4 GB) and per-part cap. Parts stay
under ~100 MB because
backend\config\models.py:275:    # free-tier edge proxies cap request bodies there (HOSTING_PLAN
$2).

```

15. assistant (2026-06-19T15:47:00.204Z)

This is the key finding. PR #204 (commit 67b6927, merged) flipped the windowing defaults ON ? `max\_window\_duration: float = 6.0` and `window\_inactivity\_end: float = 1.5` are now the SHIPPED defaults, not 0.0/OFF. The milestone is ****no longer owner-gated/default-OFF**** ? it LAUNCHED. Let me confirm the exact config field values and lines.

16. user (2026-06-19T15:47:01.132Z)

```

130         yolo_prefetch_workers: int = 2
131         min_action_window_duration: float = 2.0
132         # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md $6). 0 = OFF; > 0
= a window
133         # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
134         # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't
swallow
135         # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
136         # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13
137         # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
138         # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
139         max_window_duration: float = 6.0
140         window_min_split_gap: float = 0.5 # min internal gap (s) that qualifies as a split
point
141         # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
142         # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
143         # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
144         # Only cuts (recall can't drop). OFF by default until measured/promoted.
145         window_hard_cap: bool = False
146         # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
147         # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
148         # [?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
149         # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
150         # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant

```

```

151      # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
152      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
153      window_inactivity_end: float = 1.5
154      inactivity_rally_thresh: float = 0.20
155      inactivity_min_density: float = 0.30
156      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
157      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
158      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
159      # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
160      # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
161      # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
162      # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
163      # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
164      window_blob_fallback: bool = False
165      # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
166      # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
167      # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
168      # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
169      window_blob_factor: float = 4.0
170      log_yolo_candidates: bool = True
171      store_yolo_candidates: bool = True
172      ai_handoff_padding: float = 2.0
173      ai_max_workers: int = 5
174      # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
175      # chunks of a high-bitrate (4K) source blow past the provider upload cap
176      # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
177      # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
178      # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
179      # Matches gemini_refine's --clip-scale-width default.
180      ai_chunk_scale_width: int = 1280
181      # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
182      # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
183      # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
184      # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
185      # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
186      skip_ai_handoff: bool = False
187      # Optional debug knob: trim every video to the first N seconds before processing.
188      debug_duration: Optional[float] = None
189

```

17. assistant (2026-06-19T15:47:08.075Z)

Confirmed. The milestone (cap=6 + R4) is ****shipped as the default**** as of PR #204 (commit 67b6927, merged June 18). Now I'll cross-check each doc's claims that say the milestone is still "owner-gated / default-OFF" or "the flip is a pending next step."

Let me also confirm corpus size claims (n=11 vs n=13) and check the docs for the dates/last-updated. The reports say n=11; PR #204 says re-verified at n=13. But the quality report is a

dated checkpoint (2026-06-17) so its n=11 may be a frozen snapshot. The staleness is specifically about ****status claims**** (owner-gated / not-flipped / next-step). Let me verify the precise stale statements.

18. user (2026-06-19T15:47:11.511Z)

=== ROADMAP date 204 ===

67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)

=== was R1 launch report added ===

a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)

67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)

19. user (2026-06-19T15:47:12.662Z)

19:learns to weight ? never a hand-coded branch ? and ships ****default-OFF****, promoted only on a measured

51:| ****Promotion status**** | all cues/tiers ****default-OFF****; the milestone config flip is owner-gated |

113:All are ****default-OFF****, config-gated, and only ever ****cut**** windows (recall cannot drop):

122: with no inactivity gap to split on (a degenerate ~174 s blob). Surgical, do-no-harm, default-OFF.

297:with honest stats. Neither cue clears the bar (Table 6) ? both stay ****default-OFF****.

314:not a real defect. Adding an additive, default-OFF ****threshold-tuning** step (operating point chosen on

361: segmenter on live video is verified offline on cached trajectories, and the milestone config flip stays

362: owner-gated.

378:2. ****Promote the milestone (cap=6 + R4) to the served default**** ? a 2-line config flip, owner-gated; the

427:(default-OFF cues, bootstrap CI + paired sign test, promote-only-when-CI-clears-0, rollback = config

459: full offline mocked suite with a coverage floor. Combined with ****default-OFF** cues + a pinned CONTROL

460: variant**, rollback is a config flip, never a `git revert`.

20. assistant (2026-06-19T15:47:20.825Z)

Now I have clear evidence. The RALLY_DETECTION_QUALITY_REPORT.md is a ****dated checkpoint (2026-06-17)**** ? it precedes the flip (June 18). However, several of its status claims about the milestone being "owner-gated" / "to be promoted" are now factually contradicted by the shipped code.

But I need to be careful: the report header says "checkpoint 2026-06-17" ? a dated snapshot. The task says report CLEAR factual status staleness. The lines 51, 361-362, and 378 make present-tense status claims ("is owner-gated", "stays owner-gated", "Promote the milestone... a 2-line config flip, owner-gated") that are now false post-#204. These are the strongest candidates. However, since this is explicitly a frozen dated checkpoint, I'll be conservative and focus on the docs that are meant to be LIVE status trackers (ROADMAP, NEXT_STEPS, BACKLOG) where staleness is unambiguous, plus flag the report's now-false status lines at lower severity.

Let me check the ROADMAP and NEXT_STEPS more carefully for the milestone status, and the audio "parked" claim. Let me also verify a few other potentially-stale items. First, let me check whether Phase 3 is still "IN PROGRESS" and the audio AUC claim, and check the NEXT_STEPS milestone claim.

21. user (2026-06-19T15:47:24.264Z)

=== Gate A wiring / served path - is fusion_hybrid registered? ===

428ee44 address follow-up review on #169: correct factual error in item-3 investigation note (fusion_compare.py, fusion_golden.py, ablation.py + tests *do* exist per ls/grep on the tree; person features in fusion_hybrid + detectors/fusion_features.py; remaining M0 in-container task

is wiring into compare). Cleaned duplicate sequencing paragraph. Verified files present. Plan now accurate. M0 item 3 placement remains correct and even stronger.
5fabffd address review for #166: retitled PR (via update); restored Gate-A cheapest-win rationale, P2 FusionSegmenter specifics, served regression test finding/revert note in NEXT_STEPS; removed long P3 prose comment from fusion_hybrid.py source (moved essence to docs); confirmed archived snapshot Post-merge #165 addition is one-time finalization (no further edits to frozen archive). Fixes per owner review points. Ready for re-review.
5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step (harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline (ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real video.
6eaf85d feat(detectors): rally-start gate ? suppress between-point false fires (gate A) (#28)

=== R1 milestone doc references in NEXT_STEPS ===

a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
b593ac5 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)
c1b3733 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13

22. user (2026-06-19T15:47:25.058Z)

4:owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
23:**Also queued (owner-gated / pending input):**
54:## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)
55:**This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.
62:[Omitted long matching line]
64:- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
68:- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**;
promote to default **only on measured LOVO lift** through `run\_regression.py` (exit 0/1/3);
pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
78:**Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
126:- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next

23. assistant (2026-06-19T15:47:30.394Z)

The NEXT_STEPS.md line 378 reference was actually from the quality report. Let me look at NEXT_STEPS more precisely ? the key milestone-promotion claim. Line 54-55 describes the rally-quality track as "implementation in progress" with "all implementation owner-gated" ? but the R1 milestone flip (#204) has now shipped, and the desktop/cross-platform scope (#219) also landed. However, NEXT_STEPS is dated 2026-06-13 at the top and is a known-stale live doc. Let me focus on the clearest contradictions.

Let me re-read the NEXT_STEPS section around the milestone and the "Promote the milestone" claim, and check whether NEXT_STEPS even mentions promoting the milestone as a pending step.

24. user (2026-06-19T15:47:32.868Z)

64:- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
68:- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**;
promote to default **only on measured LOVO lift** through `run\_regression.py` (exit 0/1/3);
pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

25. user (2026-06-19T15:47:34.363Z)

docs\CODE_MAP.md:176:[Omitted long matching line]

docs\EXECUTION_POLICY.md:93: WAIT_AND_RESUME: a job raising `JobWaiting(delay\_sec, progress)` is parked (`set\_job\_waiting` ? status

docs\EXECUTION_POLICY.md:95: (`\_arm\_wake\_timer`, delay from `seconds\_until\_next\_due`, capped) and a startup drain resume parked jobs

docs\HOW_RALLY_DETECTION_WORKS.md:60: ADR-007's parked #2 and `rally\_gate.py`'s named-but-unbuilt "Gate B").

docs\archives\decisions\DECISIONS.md:295:## ADR-007: Audio "thwack" hit-detection (fused with WASB) is the next rally-detection cue; pose/stance parked

docs\archives\decisions\DECISIONS.md:361:[pose ?? stance heuristic (ours; the parked #2) ??? (feature + decision fusion)]

docs\archives\decisions\DECISIONS.md:384:(this ADR), then a stance heuristic (the parked #2), then other signals ? each cheap, debuggable, label-

docs\GOLDEN_SET_IMPLEMENTATION.md:8:[Omitted long matching line]

docs\GOLDEN_SET_IMPLEMENTATION.md:39:Each row is one PR. GS-1 ? GS-2 ? GS-4 ? GS-5 can proceed ****now****, independent of vendor selection. GS-3 is parked until you've chosen a vendor (then it's a thin adapter behind the GS-1 interface).

docs\archives\research\AUDIO_E1_FINDINGS.md:46: ***floor***. But the warm-up check (mid-match dead-time ***alone*** is ~0.6 hits/s vs ~1.3 in-rally on both clips) says

docs\MULTI_SIGNAL_FUSION_PLAN.md:41: model, with ****audio** as the #1 next cue and person/pose parked as #2******. This plan picks up the parked

docs\MULTI_SIGNAL_FUSION_PLAN.md:90:[frames ?? pose/stance (ours; parked #3) ?? stance ??????? feature + decision fusion

docs\MULTI_SIGNAL_FUSION_PLAN.md:202: detector swappable. Fine-tuning a person detector is ****parked**** ? we have no in-house person labels,

docs\MULTI_SIGNAL_FUSION_PLAN.md:212:and the features, and box detection is the cheap, robust win. ****2D** pose is parked behind the same

docs\MULTI_SIGNAL_FUSION_PLAN.md:239:| Pose (parked): ****MediaPipe Pose****, ****RTMPose / MMPose**** | Apache-2.0 | COCO-structure | ? |

docs\MULTI_SIGNAL_FUSION_PLAN.md:254:3. ****Pose now or parked**** ? ****recommend parked**** behind the interface (box-first).

docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:6:input we currently throw away. Pose/stance detection is the parked #2 (see ADR-007 / [research](RALLY_DETECTION_RESEARCH.md)).

docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:83:- Roadmap order: ****AUDIO** (now)****** ? ****[a TBD owner hunch ? slot reserved]*** ? ****POSE/STANCE** heuristic (#2, parked;

docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:104:- ? ****Not a warm-up/label artifact:**** mid-match dead time (between rallies) still has ~0.6 hits/s vs ~1.3

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:49: AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super (\$0), is ~1M params (trivially compressible later),

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:142: (PR #35) found it weak (~2.2x discrimination, AUC ~0.6) and the foreground-audio idea also failed this session; keep it

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:143: parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet first).

docs\archives\quality-iterations\README.md:20:[Omitted long matching line]

docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:51: honest LOVO): ****per-frame AUC 0.878** (Adarsh) / 0.841 (Kushagra), held-out.****** On Kushagra ? the match the

docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:59: held out and trained only on the two singles, scored AUC ****0.914**** / F1 0.729 (generalizes across play type +

docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:63:****Conclusion:**** the rally signal IS learnable from the WASB trajectory features, cross-footage (AUC ~0.85?0.91) ?

docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:145: (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now ***likely***; confirm at segment level.)

docs\QUALITY_ITERATIONS.md:723: ~+0.10****** (n=6: learned 0.583 vs heuristic 0.480), with ****AUC 0.83?0.92** decoupled from contrast******. This

docs\QUALITY_ITERATIONS.md:734: for the same reason (foreground spatial gate ****?0.04****; audio energy ****AUC ~0.6****; court-region

docs\QUALITY_ITERATIONS.md:736: ***same place and frequency band***. Only the learned model's temporal/structural features crack it (AUC

docs\QUALITY_ITERATIONS.md:744:- ****Calibration, not representation, is the small-N wall**** ? learned probabilities separate (high AUC) but

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:54:| held-out | per-frame AUC | learned F1 (real LOVO) | learned F1 (oracle threshold) | heuristic ceiling |

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:59:****The signal is learnable.**** Held-out per-frame AUC is ****0.84?0.88**** ? including on Kushagra (trained only on

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:66:> feed a learned temporal

segmenter? *Yes (AUC 0.84). The prototype is the seed of the study's Gen-0

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:80: | held-out match | heuristic LOVO | learned LOVO (AUC) | learned LOVO F1 |

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:95: from one added match (oracle ceiling 0.593, AUC 0.850 ? it will keep climbing with n).

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:100: matches, scores AUC **0.914** / F1 0.729. Different player count, camera, and frame rate, and the per-frame

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:107:- **Learned prototype (owned-model seed):** mean held-out F1 **0.615**, per-frame **AUC 0.85?0.91**, **+0.24

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:146: | held-out | per-frame AUC | learned F1 (LOVO) | oracle F1 | heuristic best |

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:168: | learned prototype | 0.377 | 0.615 | 0.568 | **0.583** (~0.64 excl. 6-rally fold) |

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:174: learned model hit **AUC 0.902 / F1 0.712** (vs heuristic 0.657) ? clean generalization to an unseen match.

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:175:- **The decoupling is the headline:** learned held-out **AUC is 0.83?0.92 on every match *regardless of contrast**, docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:192: separable by raw density, and the learned model already handles multi-court better (AUC 0.83) by combining cues.

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:197: control** (extract audio ? onset-flux + RMS energy, AUC of in-rally vs dead-time): onset/RMS AUC only **0.53?0.65**;

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:199: (Kushagra 1.1?1.7x, gBaaddy 1.3?1.9x ? confirming HF-attenuates-with-distance physics) but **AUC stays ~0.6**. **Why it

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:203: segmenter might add marginal value, but the visual learned model already hits AUC 0.85 on these videos.)

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:216: **audio energy** (AUC ~0.6), **court-region separability** (12?14% out-of-region, low headroom). They fail for the SAME

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:219: net-crossing rhythm, activity duration, spatial-spread dynamics) ? which is exactly why the prototype already hits **AUC

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:226: matches; the unsupervised heuristic extracts it on one and not the other. A learned model does (AUC 0.84).

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:229:- **Calibration, not representation, is the n=2 wall** ? learned probabilities separate (AUC 0.84) but the operating

docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md:57: | net-crossing gate | **Over-prunes this camera.** Every gate-on config caps at ~0.68; `baseline+gate2` drops recall 0.65?0.475 (loses 7 true positives). Likely net-axis estimation is off for this camera angle. |

docs\RALLY_DETECTION_QUALITY_REPORT.md:404:- **Pose / stance (parked)** ? Apache-2.0 options exist but domain shift on court angles is unproven;

docs\ROADMAP.md:13:> **Audio is parked, not "gated on golden labels":** the E1 probe and a follow-up foreground-audio test both came back

docs\ROADMAP.md:14:> weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's Gen-1 note).

docs\ROADMAP.md:87: *learned timbre classifier* ? but per the owned-model plan audio is now **parked** (the

docs\ROADMAP.md:88: foreground-audio idea also failed), **not** revived now that labels exist.

docs\ROADMAP.md:115: | + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1 probe run ? AMBER (~0.6 AUC), parked / de-prioritized |

26. assistant (2026-06-19T15:47:41.342Z)

The audio "parked" claim is consistent with the code/plan (not stale). NEXT_STEPS doesn't claim the milestone is "still to be promoted" ? it predates the milestone work entirely (dated 2026-06-13).

Now let me focus on the clearest staleness. The strongest, unambiguous status contradiction is in **RALLY_DETECTION_QUALITY_REPORT.md** ? three places say the milestone flip is "owner-gated" / "to promote" but #204 has shipped it as default. The report is a dated checkpoint, but those are present-tense status assertions that are now factually false. Let me verify the "Planned work" item 2 is the strongest one and also double-check the corpus n. Let me also verify whether the report header was meant to be updated (it says "supersedes the 2026-06-14 draft").

Let me check the git history of the quality report to understand if it's treated as a living doc

or a frozen checkpoint.

27. user (2026-06-19T15:47:44.090Z)

=== QUALITY_REPORT history ===

4bbd987 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17)

=== ROADMAP history ===

e69e23d docs: fix doc drift across 16 docs (#173)

62fb2fb docs: sync to ratified decisions + repo reality (audit truth-sync) (#71)

f9b8313 docs: fix staleness in roadmap and audio/decision docs

36044c5 docs: sync golden-set + ADRs after Roora pilot + multi-sport ingestion

f2ad8cd docs: golden-set generation + vendor process (v1), recording guidelines, ADR-006

=== NEXT_STEPS history ===

db96ddc docs: capture external due-diligence review inputs (research, pick-up-later) (#207)

e94ef62 docs: shared-data plan + next-session pickup anchor (#175)

5fabffd address review for #166: retitled PR (via update); restored Gate-A cheapest-win rationale, P2 FusionSegmenter specifics, served regression test finding/revert note in NEXT_STEPS; removed long P3 prose comment from fusion_hybrid.py source (moved essence to docs); confirmed archived snapshot Post-merge #165 addition is one-time finalization (no further edits to frozen archive). Fixes per owner review points. Ready for re-review.

5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step (harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline (ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real video.

cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)

28. assistant (2026-06-19T15:47:52.811Z)

The QUALITY_REPORT has only one commit ? it's a frozen dated checkpoint (2026-06-17). Per the task rules, I should be conservative with frozen checkpoints. However, it lives at `docs/` top level (not archived) and makes present-tense status claims that are now false. The most defensible staleness here is the "Planned work / next steps" section (§7 item 2), which lists "Promote the milestone (cap=6 + R4) to the served default" as a **pending next step** ? that work is now DONE (#204). That's a clear "completed work still listed as open."

Let me now verify the ROADMAP Phase 3 status claim. The ROADMAP says Phase 3 is "IN PROGRESS" and Phase 4 is "NEXT" ? these are roadmap framings tied to the owned-model path. I should NOT relitigate roadmap re-prioritization. But let me check: is the milestone reflected anywhere? The ROADMAP doesn't mention the windowing milestone at all, so no staleness there on that axis.

Let me verify the BACKLOG ?? items against reality ? specifically the train/eval separation guardrail (BACKLOG line 49 marks it ? "kept apart by convention only; nothing enforces"). But PR #171 (splits.py assert_disjoint) was built per the POST_166 doc. Let me verify.

29. user (2026-06-19T15:47:56.313Z)

=== splits.py exists? ===

-rw-r--r-- 1 avidu 197609 3647 Jun 15 14:11 backend/eval/splits.py

=== git log splits ===

c8ae56f address re-review for #171: fix direct-dict non-list guard hole (list(dict) keys bypass in else branch before listify); explicit dict raise matching 'must be a list' test msg. Verified smoke (direct now raises, paths preserved, happy/strict/leakage OK). plan append-only re-review entry + gate note. (per owner re-review comment 4702451231; fixes the CI red test_load_split_non_list_raises). All green where executable in container (owner re-runs full). No self-merge.

e1b6e68 address review for #171: non-list raise+check; explicit match key raise (no name/video fallback); strict train/eval split; pathlib+utf8sig; added leakage tests (same match diff vid, end-to-end, non-list, falsy, bad label) with comments. plan append-only review entry. (fixes 1-4 must, 5-6 nice)

ebdf331 fix: CI for PR171 - wrap long docstring lines (E501), use cast for json.load mypy
compat, ensure files end with newline
e79bbc6 feat: M0 train/eval guard core - splits.py assert_disjoint + load_split +
tests/test_splits.py (per plan M0 + BACKLOG/DEFERRED #13)

=== is it wired into run_regression / training? ===
backend/eval/splits.py
backend/eval/___pycache___/splits.cpython-313.pyc

30. assistant (2026-06-19T15:48:07.557Z)

So ``backend/eval/splits.py::assert_disjoint`` exists (PR #171, merged), but it is ****NOT** wired into `run_regression.py` or the training entrypoint ? it's only the pure-logic core + tests. Let me re-read the BACKLOG entry (line 49) carefully. It says the guardrail's "Concrete design ... new ``backend/eval/splits.py`` with ``load_split(...)`` ... + ``assert_disjoint(...)`` ... wire the guard into the training entrypoint ... and into ``run_regression.py``".

The BACKLOG marks this ? with *****Revisit:***** dedicated design session with the user (needs a collector-side manifest change) ? implying NOTHING is built. But the pure-logic core (``splits.py`` + ``assert_disjoint`` + ``load_split`` + unit tests) IS built and merged (#171). The wiring into training/run_regression and the collector manifest field are deferred. So the BACKLOG entry is ****partially stale****: it presents the whole item as not-started ("nothing enforces", revisit = design session) when the splits.py module + assert_disjoint + leakage unit tests are already built.

This is exactly the kind of "in fact built and merged" staleness. But I must be careful ? the BACKLOG item describes a multi-part deliverable, and the WIRING (the part that actually enforces in the pipeline) is genuinely not done. The ? "nothing enforces that a video used to train... is absent from the eval/regression set" is still TRUE at the pipeline level (splits.py isn't called by training/run_regression). So the core CLAIM (nothing enforces) is technically still accurate. I'll flag the narrower factual sub-claim ? that splits.py and assert_disjoint don't yet exist ? as stale, since the BACKLOG describes them as future "Concrete design" to build.

Let me look at the exact BACKLOG wording again to craft a precise, minimal edit.

31. user (2026-06-19T15:48:08.547Z)

49 - ? ****Train/eval separation guardrails**** (was ``DEFERRED_HARDENING_PLAN`` #13; design preserved here on archival 2026-06-14) ? train & eval videos live in ****separate catalogs**** today, kept apart by convention only; nothing ***enforces*** that a video used to train the learned segmenter (``backend/eval/training.py::build_training_set`` ? ``distill_local``) is absent from the eval/regression set, so leakage would silently inflate the headline metrics. ****Concrete design**** (from the archived DEFERRED plan §2): split unit = ****match, never clip**** (hold ~? of each sport as eval, per ``GOLDEN_SET_PHASE1_VIDEOS.md``); add a ``split: "train"|"eval"`` field to the collector's ``curate export`` manifest (cross-repo contract, ADR-002); new ``backend/eval/splits.py`` with ``load_split(manifest)`` -> (train_ids, eval_ids) keyed by ****match id**** (not file MD5 ? re-encodes share a match but differ in ``compute_video_id``) + ``assert_disjoint(train_ids, eval_ids)`` that raises on overlap; wire the guard into the training entrypoint (``distill_local`/`training``) and into ``run_regression.py`` (baseline computed strictly on the eval split). ****Verification:**** unit test (overlapping ids ? ``assert_disjoint`` raises) + integration (a deliberately-leaked manifest fails the training entrypoint). Full rationale/risks: [``archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md``](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2. ****Revisit:**** dedicated design session with the user (needs a collector-side manifest change).

32. assistant (2026-06-19T15:48:17.908Z)

This BACKLOG entry describes ``backend/eval/splits.py`` with ``load_split`` and ``assert_disjoint`` as future "Concrete design" to build, but they are already BUILT and merged (#171, commit e79bbc6, file ``backend/eval/splits.py`` present). The remaining work is the wiring + the collector manifest field. I'll craft a surgical edit that marks the pure-logic core as DONE while preserving the still-open wiring. This is medium severity (partially built; core merged, wiring deferred).

Now let me verify the BACKLOG "? `run\_all\_tests.py` added at repo root" and a couple other ? claims to make sure I'm not missing reverse-staleness (completed marked open / open marked done). Let me also double check the REAL_FOOTAGE report status header. The REAL_FOOTAGE report is dated 2026-06-16 with "Status: validation complete, roadmap proposed (all flags default-OFF)". Its R1 lever (line 80) proposes "Ship the smallest safe milestone" ? that has now shipped via #204. But again, that report is a dated proposal/roadmap doc. Let me check its history and status header treatment.

33. user (2026-06-19T15:48:18.743Z)

=== REAL_FOOTAGE history ===

c483346 docs(quality): real-footage validation capstone + first ground-truth at-par + dual-eval-set discipline (#196)

=== POST_166 history ===

c8ae56f address re-review for #171: fix direct-dict non-list guard hole (list(dict) keys bypass in else branch before listify); explicit dict raise matching 'must be a list' test msg. Verified smoke (direct now raises, paths preserved, happy/strict/leakage OK). plan append-only re-review entry + gate note. (per owner re-review comment 4702451231; fixes the CI red test_load_split_non_list_raises). All green where executable in container (owner re-runs full). No self-merge.

e1b6e68 address review for #171: non-list raise+check; explicit match key raise (no name/video fallback); strict train/eval split; pathlib+utf8sig; added leakage tests (same match diff vid, end-to-end, non-list, falsy, bad label) with comments. plan append-only review entry. (fixes 1-4 must, 5-6 nice)

e79bbc6 feat: M0 train/eval guard core - splits.py assert_disjoint + load_split + tests/test_splits.py (per plan M0 + BACKLOG/DEFERRED #13)

f039976 address review for #170: provisional blocker 'NOT YET CALIBRATED' (replaces UNDEFINED); reword docstring/comments no 'ratified' (provisional, #172); update test assert + #172 + stale comment. plan note + review round entry updated. (per owner: honest + green test)

99ad258 feat: M0 quick win #1 - ratify ship-gate target + clear UNDEFINED blocker (per POST_166 plan)

428ee44 address follow-up review on #169: correct factual error in item-3 investigation note (fusion_compare.py, fusion_golden.py, ablation.py + tests *do* exist per ls/grep on the tree; person features in fusion_hybrid + detectors/fusion_features.py; remaining M0 in-container task is wiring into compare). Cleaned duplicate sequencing paragraph. Verified files present. Plan now accurate. M0 item 3 placement remains correct and even stronger.

b649f1c address review on #169: revert frozen archive edit (should-fix); trim handoff to short pointer per #165 intent (record moved to plan); add Transition Record section; unbundle catalog items 3/6/10 + clarify P3 'shipped' state with investigation (no fusion_compare.py etc. present; general harness + P2 are); add M0 Quick Wins (the 4 in-container items per course-correction); re-sequence milestones (M0 quick wins first + owner-led parallel, heavy last); update risk table + sequencing text. Thorough point-by-point in PR comment.

1b2450a docs: finalize handoff/audit after #166 merge + introduce post-166 work items plan with risk assessment, milestones and sequencing (review first per owner)

34. assistant (2026-06-19T15:48:29.525Z)

The POST_166 doc is an append-only project plan/log (it contains its own dated execution records, including the #171 splits guard "executed" entries at lines 161-196). It's a tracked project doc with its own internal log, so its forward-looking milestones are framed as a plan ? I should NOT treat planned milestones in a frozen plan doc as staleness (that's roadmap re-prioritization). Its internal log already records #171 as done. So POST_166 has no clear status staleness to fix.

The REAL_FOOTAGE report is a single-commit dated capstone (2026-06-16), status "validation complete, roadmap proposed (all flags default-OFF)". Its R1 proposal predates the flip. Like the quality report, it's a frozen dated report; flagging its forward roadmap as stale would be roadmap re-prioritization. I'll leave its proposal section alone but note: its header says "all flags default-OFF" as a present-tense status, which is now false for the served default. However, that header describes the state at the time of the report (the proposal was for flags to ship OFF). I'll be conservative and not touch the frozen dated reports' proposals.

Let me now make the strongest, most defensible determination. The clearest, unambiguous code-contradicting staleness is in the ****RALLY_DETECTION_QUALITY_REPORT.md**** "Promotion status"

summary row and §7 planned-work item ? these are present-tense status claims now contradicted by shipped code (#204). Even though it's a dated checkpoint, it lives at top-level `docs/` and the task explicitly targets this file. Let me re-read those exact lines to craft minimal edits, and also re-examine the abstract's headline which says "the milestone config flip is owner-gated."

35. user (2026-06-19T15:48:31.000Z)

```
355      - **Sample size.** n=11 (5 with real ground truth) ? enough for direction-checks, LOVO
sign-tests, and
356      now significant windowing results, but not enough to claim a general model. Corpus
growth is the
357      highest-ROI unblock.
358      - **Person & audio CIs span 0** at n=6; neither fusion cue is promotable yet.
359      - **Court polygons absent** ? the mechanical reason +person is a wash (counts
spectators).
360      - **Served path partially live-verified.** The over-merge windowing is served-measured;
the full fusion
361      segmenter on live video is verified offline on cached trajectories, and the milestone
config flip stays
362      owner-gated.
363      - **Detector domain gap unmeasured.** WASB/TrackNet and the person detector were trained
on other rigs;
364      their accuracy on our amateur/static-tripod footage is not yet quantified, and every
downstream cue
365      inherits that gap.
366      - **C3 ? WASB checkpoint provenance** is a commercial-ship blocker a clean head does not
launder; an
367      owned or cleanly-licensed detector is required before commercial release.
368
369      ---
370
371      ## 7. Planned work ? next steps & expected wins
372
373      1. **Grow the corpus via the oracle (the #1 lever).** Priority: multi-court badminton
(the target
374      distribution), then ?3 matches per new sport (table tennis first ? WASB transfers at
F1 0.909 in the
375      literature). **Expected win:** at n=10 with 9/10 wins the exact sign-test reaches **p
? 0.011** ? and
376      we are already at n=11 with 10/0 on the milestone (p=0.002), so the cue-level results
become
377      publishable as the corpus and ablations widen.
378      2. **Promote the milestone (cap=6 + R4) to the served default** ? a 2-line config flip,
owner-gated; the
379      significant, served-measured, no-regression win of this cycle.
380      3. **Close the rally-END gap** ? the ground-truth-localized lever (|?end| 3.8 s ? toward
Gemini's ~1 s):
```

36. assistant (2026-06-19T15:48:42.800Z)

Now I have all the evidence I need. Let me make my determinations carefully.

The RALLY_DETECTION_QUALITY_REPORT.md is a **dated technical-paper checkpoint (2026-06-17)** with a single commit. It is genuinely a frozen snapshot of measured results at that date. The task rules say to be conservative and not relitigate; a dated checkpoint's measured tables (n=11, etc.) are legitimately frozen-in-time and should NOT be "corrected" to n=13.

HOWEVER, two specific present-tense **status** claims in it are now factually contradicted by shipped code and are not "measured results" but live status assertions:

- Line 51 (Promotion status row): "the milestone config flip is **owner-gated**"
- Line 361-362 (Limitations): "the milestone config flip **stays owner-gated**"
- Line 378 (§7 item 2): "***Promote the milestone (cap=6 + R4) to the served default** ? a 2-line config flip, owner-gated" ? listed as planned/pending work.

All three are contradicted by PR #204 (commit 67b6927, merged): `max_window_duration: float =

6.0` and `window_inactivity_end: float = 1.5` are the SHIPPED defaults. The milestone is no longer owner-gated; it has been flipped to the served default.

Given the file is a dated checkpoint, I'll flag these as the staleness items but keep severity calibrated (medium) and make minimal surgical edits that note the flip shipped (#204) without rewriting the measured tables. The \$7 "planned work" item 2 (completed work listed as open) is the clearest "high" ? it presents shipped work as a next step.

Let me verify one more time that no doc in my slice has already been updated to reflect the flip, and confirm the BACKLOG splits item is the only build-status one. Let me check the QUALITY_ITERATIONS reference and confirm the "R1 LAUNCHED" entry exists (to prove the flip is real and recorded), then finalize.

37. user (2026-06-19T15:48:45.018Z)

```
104:### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13
*(2026-06-18)*
107:the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report
convention).
112:| transition | n=11 (filed) | **n=13 (launched)** |
209:shipped behaviour, so it is NOT flipped here. (W2 net-crossings stays default-OFF + retained
? no GT-backed lift.)
248: candidate *before* it's flipped ON). Frozen numbers (n=11, this commit): DEFAULT tolF1
**0.091** / f1@0.5 0.194;
423:them.) Kept **default-OFF** pending owner go to flip the served default (one config line);
the residual over-split is the
555:**The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.** A
served-side check ?
603:candidate set; flipping the **live served default** stayed owner-gated at the time of this
entry. **Update: the
```

38. assistant (2026-06-19T15:48:49.773Z)

Confirmed: QUALITY_ITERATIONS.md line 104 records "R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON)" as of 2026-06-18 (#204). This conclusively proves the milestone is shipped, not owner-gated. The RALLY_DETECTION_QUALITY_REPORT.md (dated 2026-06-17, before the flip) is now stale on those three status claims.

I have enough to finalize. Let me prepare the verbatim stale_text for each finding. Let me grab the exact text of line 51 and the abstract reference.

39. user (2026-06-19T15:48:50.842Z)

```
50      | **Golden corpus** | **n=11 matches** (grown from 6), incl. 5 real ground-truth clips;
minors excluded; no biometric labelling |
51      | **Promotion status** | all cues/tiers **default-OFF**; the milestone config flip is
owner-gated |
```

40. assistant (2026-06-19T15:49:00.133Z)

I now have all the verbatim text. Let me finalize my findings.

Summary of verified staleness in my slice (all contradicted by PR #204 / commit 67b6927, merged 2026-06-18, which flipped `max\_window\_duration: float = 6.0` and `window\_inactivity\_end: float = 1.5` as the shipped defaults in `backend/config/models.py:139,153`; recorded as "R1 ? MILESTONE LAUNCHED" in `docs/QUALITY\_ITERATIONS.md:104`):

1. **`docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:378`** (\$7 Planned work item 2) ? lists "Promote the milestone (cap=6 + R4) to the served default" as pending next work; it has shipped. HIGH (completed work listed as open).
2. **`docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:51`** (Promotion-status row) ? "the milestone config flip is owner-gated"; no longer gated/shipped. MEDIUM.
3. **`docs/RALLY\_DETECTION\_QUALITY\_REPORT.md:361-362`** (Limitations) ? "the milestone config flip stays owner-gated"; shipped. MEDIUM.

4. `**docs/BACKLOG.md:49**` (Train/eval separation guardrails) ? describes ``backend/eval/splits.py`` with ``load_split`/`assert_disjoint`` + unit test as future "Concrete design" to build; they are already built and merged (#171, file present at ``backend/eval/splits.py``). The wiring + collector manifest field remain genuinely deferred. MEDIUM (partially built; core merged).

I'll keep the RALLY_DETECTION_QUALITY_REPORT edits minimal and append the launch note rather than rewriting frozen tables, since it's a dated checkpoint.

41. user (2026-06-19T15:49:18.897Z)

Structured output provided successfully

42. assistant (2026-06-19T15:49:31.064Z)

I have completed the audit of my slice and returned the verified staleness findings via StructuredOutput.

Summary of what I found (all in docs that are status/roadmap trackers, contradicted by shipped code):

`**Root cause ? PR #204 (commit 67b6927, merged 2026-06-18)**` flipped the windowing milestone defaults ON in ``backend/config/models.py``: ``max_window_duration: float = 6.0`` (line 139) and ``window_inactivity_end: float = 1.5`` (line 153), recorded as "R1 ? MILESTONE LAUNCHED" in ``docs/QUALITY_ITERATIONS.md:104``. Three claims in ``docs/RALLY_DETECTION_QUALITY_REPORT.md`` (a 2026-06-17 checkpoint, one day before the flip) still describe that flip as owner-gated / pending:

- ``:378`` (\$7 Planned-work item 2) lists "Promote the milestone (cap=6 + R4) to the served default" as a next step ? high severity (completed work listed as open).
- ``:51`` (Promotion-status row) says "the milestone config flip is owner-gated."
- ``:361-362`` (Limitations) says "the milestone config flip stays owner-gated."

`**Separately, `docs/BACKLOG.md:49**`` describes ``backend/eval/splits.py`` (``load_split`/`assert_disjoint`` + unit test) as future "Concrete design" to build, but the pure-logic core was built and merged in PR #171 (commit e79bbc6) ? ``backend/eval/splits.py`` and ``tests/test_splits.py`` are present. The pipeline wiring and the collector manifest ``split`` field remain genuinely unbuilt, so I scoped the correction to only mark the merged core as done.

I checked but did NOT flag:

- ``docs/ROADMAP.md`` Phase 3/4 framing and the "audio parked (~0.6 AUC)" note ? consistent with the owned-model plan and code; not stale.
- ``docs/NEXT_STEPS.md`` ? dated 2026-06-13, predates the milestone work; its "owner-gated / implementation in progress" framing is roadmap state, not a code-contradicted status claim, and it never lists the milestone-promotion as a pending step.
- ``docs/POST_166_...md`` ? an append-only project plan whose own internal log already records #171 as executed; its forward milestones are a plan, not staleness.
- ``docs/REAL_FOOTAGE_VALIDATION_REPORT.md`` ? a single-commit dated capstone (2026-06-16) whose R1 "ship the smallest safe milestone" is a proposal; treating that as stale would be roadmap re-prioritization, which is out of scope.